

Practice booklet for HKDSE Compulsory Mathematics

Topics included: Counting, Percentage, and Probability

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We express our deep respect to all past educators, exam boards, and authors whose contributions have shaped the foundation of mathematical excellence in Hong Kong.

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1 Counting

1.1 The Law of Addition

“Before we can sum things up, our minds must see them as separate pools. Categorization is not just sorting; it is the fundamental rule that allows arithmetic to safely begin without anything overlapping.”

Definition: The Law of Addition

When a total collection is split into completely distinct, non-overlapping categories, the total count of items across all categories is equal to the sum of the counts of each individual category. If category A has m items and category B has n items, and they share no items, there are $m + n$ total choices.

Intuitive Core: We use addition when our problem can be split using the word “OR”. If you choose an item from category A *or* from category B , you simply add their totals. **WARNING: NEVER ADD COUNTS IF AN ITEM CAN BELONG TO BOTH CATEGORIES AT THE SAME TIME!**

Example: Categorization by Adding

A student wants to buy a beverage. A kiosk sells 3 distinct brands of fruit juice and 4 distinct brands of sodas. How many total choices does the student have?

Detailed Solution Steps:

Let J be the category of juices, and S be the category of sodas.

Count of choices in $J = 3$

Count of choices in $S = 4$

Since a drink cannot be *both* a fruit juice and a soda simultaneously, these categories are completely non-overlapping.

Total choices = $3 + 4 = 7$

Hint / Concept Review

Identify the keyword *or*. The student can choose a juice **OR** a soda.

Check for DOUBLE COUNTING: Can a drink be a juice and soda at once? No. Therefore, apply direct addition.

Exercise 1.1.1

A set contains numbers from separate pools: Pool $X = \{2, 4, 6\}$ and Pool $Y = \{1, 3, 5, 7\}$. If a single number is selected from either Pool X *or* Pool Y , how many total possible outcomes are there?

- A. 3
- B. 4
- C. 7
- D. 12

Scratch Space:

Exercise 1.1.2

An alphabet set contains 5 distinct vowels and 21 distinct consonants.

- (a) State the condition required to apply the Law of Addition to find the total number of single-letter choices.
- (b) Find the total number of unique single-letter outcomes.

Solution Space:

Exercise 1.1.3

A local bakery offers 8 varieties of savory breads and 11 varieties of sweet pastries. If Mr. Mok wants to buy exactly one bakery item for breakfast, how many unique options can he select?

- A. 3
- B. 8
- C. 11
- D. 19

Scratch Space:

Exercise 1.1.4

Owen goes to a stationer's shop. There are 14 different black roller pens and 9 different blue gel pens on display.

- (a) Write down an arithmetic expression matching the total choices of picking one pen.
- (b) Calculate the total number of pen choices.

Solution Space:

Exercise 1.1.5

Category A contains x distinct objects, and Category B contains $2x + 3$ distinct objects. Given that Category A and Category B share zero common elements, express the total number of ways to pick one object from either category in terms of x .

- A. $x + 3$
- B. $2x + 3$
- C. $3x + 3$
- D. $2x^2 + 3x$

Scratch Space:

Exercise 1.1.6

A locker contains n standard math rulers and $3n - 1$ geometric compasses.

- (a) Find a simplified expression for the total items in the locker.
- (b) If the total number of single item options is 39, determine the value of n .

Solution Space:

Exercise 1.1.7

A class committee consists of 4 teachers from School A and 5 teachers from School B [cite: 2]. If only one teacher is randomly selected to lead a district seminar, find the number of possible outcomes.

- A. 1
- B. 4
- C. 5
- D. 9

Scratch Space:

Exercise 1.1.8

In a recreation club, there are 75 male members and 105 female members[cite: 3]. A single coordinator needs to be chosen from the club members.

- (a) Evaluate the total number of choices available to fill the coordinator position.
- (b) TRAP WARNING: If 5 male members are also registered under an overlapping duplicate sub-list, explain why direct addition of the full unverified lists fails.

Solution Space:

Exercise 1.1.9

An integer is to be chosen from the set $\{1, 2, 3, \dots, 20\}$. Let category M be the multiples of 7, and category N be the multiples of 13. How many options exist to choose a number belonging to M or N ?

- A. 1
- B. 2
- C. 3
- D. 4

Scratch Space:

Exercise 1.1.10

A raffle box holds tickets marked with unique identity integers. There are k premium tickets and $k^2 - 2$ basic tickets, with absolutely no tickets sharing categories.

- (a) Write down a polynomial expression for the total choices of pulling out one ticket.
- (b) Given that the single-ticket pool count equals 70, find the value of k .

Solution Space:

Exercise 1.1.11

Suppose a student counts the cards in a deck. There are 26 red cards and 12 face cards. If the student calculates total unique choices of picking a single card that is red *or* a face card by computing $26 + 12 = 38$, this calculation is incorrect because:

- A. Addition cannot combine cards.
- B. Some face cards are also red, causing overlap.
- C. The numbers should be multiplied.
- D. Red cards cannot be counted with face cards.

Scratch Space:

Exercise 1.1.12

A teacher reviews an error made by a student trying to combine overlapping sets. Group A consists of students who study History, and Group B consists of students who study Geography. Some students study both subjects.

- (a) State why the expression $\text{Total} = \text{Count}(A) + \text{Count}(B)$ overestimates the true collection pool.
- (b) Formulate the corrections needed conceptually when elements break the distinct category rule.

Solution Space:

1.2 The Law of Subtraction

“When paths cross, the landscape gets counted twice. Subtraction isn’t just about taking things away; it’s our eraser for clearing out the accidental clones we create when sets overlap.”

Theorem: The Law of Subtraction (Inclusion-Exclusion Principle)

If two counting pools, Category A and Category B , share a set of common items that belong to *both* pools simultaneously, adding their raw counts will double-count those shared items. The true total number of unique items belonging to Category A or Category B is given by:

$$\text{Total} = \text{Count}(A) + \text{Count}(B) - \text{Count}(A \text{ and } B)$$

Intuitive Core: When lists overlap, direct addition breaks down. We must count everything inside Category A , add everything inside Category B , and then **subtract the shared elements exactly once** to undo the double-counting. **WARNING: ALWAYS IDENTIFY AND REMOVE THE DUPLICATE OVERLAP COUNT BEFORE DECLARING A FINAL TOTAL!**

Example: Removing Duplicacy from Overlapping Pools

In a small workshop of 20 students, 12 students sign up for a Mathematics module and 10 students sign up for an Art module. If 4 students are found to have signed up for *both* modules, how many unique students have signed up for at least one module?

Detailed Solution Steps:

Let M be the set of Math students, and A be the set of Art students.

$$\text{Count}(M) = 12$$

$$\text{Count}(A) = 10$$

The overlapping double-counted category is $\text{Count}(M \text{ and } A) = 4$.

Using the Law of Subtraction:

$$\text{Total} = 12 + 10 - 4$$

$$\text{Total} = 22 - 4 = 18$$

Hint / Concept Review

Notice that $12 + 10 = 22$, which exceeds the total population of 20 students! This is a clear signal that *duplicacy* exists.

By subtracting the 4 overlapping students who were counted once in Math and once in Art, we recover the true unique count.

Exercise 1.2.1

List X contains 15 items, and List Y contains 22 items. If there are exactly 6 items that appear on both lists simultaneously, how many unique items are there in total across both lists?

- A. 31
- B. 37
- C. 43
- D. 49

Scratch Space:

Exercise 1.2.2

Consider two mathematical data sets, Set A and Set B . It is given that the number of elements inside A is 45, the number of elements inside B is 30, and they share 12 elements in common.

- (a) Write out the formulation needed to extract the combined unique total without double-counting.
- (b) Calculate the total count of individual elements.

Solution Space:

Exercise 1.2.3

In a survey of a class with 35 children, 20 say they like playing football and 18 say they like badminton. If 8 children enjoy both sports, how many unique children enjoy at least one of these sports?

- A. 22
- B. 30
- C. 35
- D. 38

Scratch Space:

Exercise 1.2.4

Owen's school tracking records indicate that 65 students joined the swimming gala and 40 students joined the track events. However, a verification reveals that 15 of these athletes participated in both competitive tracks.

- (a) Find the total number of individual students who participated in at least one event.
- (b) If the school has 100 students in total, deduce how many students did not participate in either event.

Solution Space:

Exercise 1.2.5

Category P has $3k$ items, and Category Q has $5k - 2$ items. If they share exactly $k + 1$ overlapping items, find an expression for the total count of unique elements in terms of k .

- A. $7k - 3$
- B. $7k - 1$
- C. $8k - 3$
- D. $9k - 1$

Scratch Space:

Exercise 1.2.6

An automated library system categorizes books dynamically. Vault X holds $n^2 + 5$ records, Vault Y holds $8n - 2$ records, and the internal tracking loop shows an overlap of $2n + 1$ duplicate entries.

- (a) Construct a simplified polynomial expression representing the true total number of distinct book records.
- (b) Given that the actual total count of unique records is 67, determine the positive integer value of n .

Solution Space:

Exercise 1.2.7

In a recruitment exercise, the total number of male applicants is 140 and the number of female applicants is 110. If 30 applicants hold an overlapping duplicate registration status across both sub-lists, how many total unique individual candidates are registered in the tracking database?[cite: 3]

- A. 220
- B. 250
- C. 280
- D. 300

Scratch Space:

Exercise 1.2.8

A local community council runs an independent review of candidate databases. There are 180 total community members registered[cite: 3]. List A tracks senior hobbyists and contains 115 entries. List B tracks local volunteers and contains 95 entries.

- (a) If every member is on at least one list, compute the number of elements contained in the overlapping duplicate pool.
- (b) Formulate an explicit structural breakdown proving how many members belong exclusively to List A .

Solution Space:

Exercise 1.2.9

An integer is randomly drawn from the uniform set $\{1, 2, 3, \dots, 30\}$. Let Category A be the multiples of 2, and Category B be the multiples of 3. How many unique numbers from the set belong to Category A *or* Category B ?

- A. 15
- B. 20
- C. 25
- D. 30

Scratch Space:

Exercise 1.2.10

A factory tracks components using a dual-line checking procedure. Line 1 flags items with physical blemishes, processing $m^2 - m$ elements. Line 2 flags structural calibration shifts, processing $10m + 4$ elements. The diagnostic report shows that $3m - 1$ elements triggered both lines.

- (a) State the simplified expression for the total unique flagged components.
- (b) If the distinct count of bad components is 85, solve for the structural metric parameter m .

Solution Space:

Exercise 1.2.11

A student attempts to find the total items across two overlapping pools by writing down: $\text{Total} = \text{Count}(A) + \text{Count}(B) + \text{Count}(A \text{ and } B)$. This calculation fails because:

- A. The overlapping items are completely ignored.
- B. The overlapping items are triple-counted instead of removed.
- C. Multiplication should be used instead of addition.
- D. Subtraction cannot fix overlapping arrays.

Scratch Space:

Exercise 1.2.12

An analyst evaluates two raw registration systems for a major public exam. System A records 1200 entries, and System B records 1500 entries. The analyst simply prints the sum as 2700 entries.

- (a) Discuss the critical conditions under which this evaluation fundamentally fails.
- (b) Frame the necessary administrative correction steps required if 350 applicants submitted duplicate forms across both capture networks.

Solution Space:

1.3 The Law of Multiplication

“When choices cascade one after another, a simple list explodes into a grid. Multiplication is nothing more than tracking how a single choice branches out into a landscape of secondary options.”

Theorem: The Law of Multiplication (The Product Rule)

If a compound task can be broken down into two consecutive, independent stages, where Stage 1 can be completed in m different ways, and Stage 2 can be completed in n different ways regardless of the outcome of Stage 1, then the entire multi-stage task can be completed in exactly $m \times n$ unique ways.

Intuitive Core: We use multiplication when our structural stages are combined using the word “AND”. If you must pick an item from category A *and* then a subsequent item from category B , your possibilities form a grid structure, transforming counting into direct multiplication. **TRAP WARNING: THE NUMBER OF CHOICES IN THE SECOND STAGE MUST REMAIN CONSTANT REGARDLESS OF WHICH SPECIFIC OPTION WAS PICKED IN THE FIRST STAGE!**

Example: Tabulating Combinations via Grids

A student chooses a lunch combo consisting of 1 main dish and 1 drink. The canteen offers 4 distinct main dishes and 3 distinct drinks. Find the total number of unique lunch combinations available.

Detailed Solution Steps:

Let Stage 1 be choosing the main dish.

Ways to complete Stage 1 (m) = 4

Let Stage 2 be choosing the beverage.

Ways to complete Stage 2 (n) = 3

Since a complete combo requires a main dish *and* a drink, the total structural options form a 4×3 grid.

Total Combinations = $4 \times 3 = 12$

Hint / Concept Review

Think of this as a grid where the 4 main dishes head the rows, and the 3 drinks head the columns.

Every intersecting cell represents an individual combo. Instead of counting all 12 cells manually, basic arithmetic gives $4 \times 3 = 12$.

Subsection 1.3 Practice Exercises**Exercise 1.3.1**

An alphanumeric code is formed by combining a single letter chosen from the set $\{A, B, C\}$ followed by a single digit chosen from the set $\{1, 2, 3, 4, 5\}$. How many unique two-character codes can be generated?

- A. 8
- B. 12
- C. 15
- D. 20

Scratch Space:

Exercise 1.3.2

A game token tracker registers states using two variable dials. Dial 1 contains 6 discrete symbols, and Dial 2 contains 8 discrete symbols.

- (a) Explain why the total variation profile is evaluated through a multiplication calculation instead of basic addition.
- (b) Calculate the total number of unique structural token states.

Solution Space:

Exercise 1.3.3

Mr. Mok is putting together an outfit. He has 5 clean shirts and 6 distinct trousers to choose from. If he chooses exactly one shirt *and* one pair of trousers, how many unique outfit pairings can he wear?

- A. 11
- B. 25
- C. 30
- D. 36

Scratch Space:

Exercise 1.3.4

A smart-lock device requires a user to input a primary security color code followed by a secondary numeric button pass. There are 3 color options available and 12 numeric key buttons.

- (a) Set up a two-stage multiplication map representing the total unique passcodes.
- (b) Find the total number of unique security passcode options.

Solution Space:

Exercise 1.3.5

An experiment requires running consecutive trials. Stage A yields $x + 2$ distinct configurations, and Stage B yields $2x$ distinct outcomes. Express the total number of compound outcome profiles in terms of x .

- A. $3x + 2$
- B. $2x^2 + 2$
- C. $2x^2 + 4x$
- D. $4x^2 + 2x$

Scratch Space:

Exercise 1.3.6

A systematic manufacturing line encodes microchips using a row-by-column lattice design. The matrix contains n horizontal rows and $2n + 5$ vertical tracks.

- (a) Construct a simplified algebraic expression representing the total coordinate intersection pin options on a chip.
- (b) If the total number of unique coordinate pins evaluates to 52, find the integer parameter n .

Solution Space:

Exercise 1.3.7

A professional qualification exam format consists of a theory paper split into multiple choices and a practical laboratory assignment[cite: 1]. If there are 8 distinct theory topic sets and 5 distinct lab stations, how many different ways can a candidate be assigned an exam sequence?

- A. 13
- B. 40
- C. 56
- D. 64

Scratch Space:

Exercise 1.3.8

An automated vehicle licensing framework processes license plates sequentially using registration parameters. A basic code sequence requires selecting 1 school identifier from 5 choices, followed by 1 area category code out of 16 alternatives[cite: 4].

- (a) Find the total number of unique license code configurations that can be assigned.
- (b) TRAP WARNING: Suppose 2 of the area category codes are discontinued exclusively whenever the first school identifier is selected. Explain why a single multiplication step over the unadjusted boundaries fails.

Solution Space:

Exercise 1.3.9

Two fair dice—one colored blue and one colored green—are tossed once[cite: 4]. How many total unique sample space element pairings (b, g) can be logged in the outcome registry?

- A. 12
- B. 18
- C. 36
- D. 40

Scratch Space:

Exercise 1.3.10

A specialized lottery generator selects a primary integer key followed by a secondary alphanumeric validation index. The primary key is chosen from the set $\{1, 2, \dots, p\}$, and the secondary index is chosen from a collection of size $p + 3$.

- (a) Formulate the structural polynomial tracking the complete system passcode configuration space.
- (b) If the total configuration size is measured to be 40, determine the structural parameter p .

Solution Space:

Exercise 1.3.11

A student is told to find the total number of two-letter pairs using letters from a set of 10 vowels and 10 consonants, where no letter can be used twice. The student incorrectly calculates the total as $10 \times 10 = 100$ combinations for choosing two vowels. This calculation fails because:

- A. Stage boundaries are entirely non-numeric.
- B. The choice pool for the second stage drops because replacement is prohibited, rendering independent multiplication invalid.
- C. The counts should have been added instead.
- D. Vowels cannot form grids.

Scratch Space:

Exercise 1.3.12

An engineer designs an access system using a two-stage sequential path. Stage 1 presents 5 distinct pathways. If a user selects Path 1, Stage 2 offers 4 onward nodes. If a user selects any other path in Stage 1, Stage 2 offers 7 onward nodes.

- (a) Analyze why computing the total combinations as $5 \times 7 = 35$ introduces a systemic error.
- (b) Correct the structure by deploying a breakdown using both the Law of Addition and the Law of Multiplication.

Solution Space:

1.4 The Venn Diagram

“When boundaries blur across multiple overlapping worlds, simple math equations can become overwhelming. The Venn diagram maps these complex logical intersections into visible, spatial boundaries that our minds can instantly grasp.”

Definition: The Venn Diagram

A Venn diagram is a geometric representation of sets where a bounding rectangle depicts the *universal set* (the entire population under discussion), and overlapping closed regions (usually circles) represent distinct tracking categories. The areas created by these intersections allow sorting based on combined properties.

Intuitive Core: When managing more than 2 interacting categories, listing combinations sequentially gets messy. Drawing overlapping circles helps partition our items into mutually exclusive regions. **WARNING: ALWAYS FILL THE VENN DIAGRAM FROM THE INNERMOST INTERSECTION OUTWARD TO AVOID DOUBLE COUNTING REPEATED SUBSETS!**

Example: Sorting Three Intersecting Groups Visualized

A youth center surveys 40 students regarding their extracurricular hobbies: 22 play Football (F), 18 play Badminton (B), and 7 participate in both ($F \cap B$). Map this distribution using a spatial grid.

Detailed Solution Steps:

1. Start with the innermost intersection region:

$$\text{Count}(F \text{ and } B) = 7$$

2. Compute the exclusive region for F by stripping out the shared core:

$$\text{Exclusive } F = 22 - 7 = 15$$

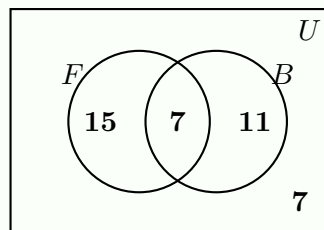
3. Compute the exclusive region for B by stripping out the shared core:

$$\text{Exclusive } B = 18 - 7 = 11$$

4. Sum all interior elements to identify non-participants:

$$\text{Total active} = 15 + 7 + 11 = 33$$

$$\text{Outside Circle Pool} = 40 - 33 = 7$$

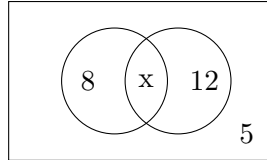


Hint / Concept Review

Always lock down the middle intersection (**7**) first. If you write 22 directly in the left circle, you double-count the overlapping core!

Subsection 1.4 Practice Exercises**Exercise 1.4.1**

The tracking diagram below displays items categorized into sets P and Q .



If the total number of unique elements inside the universal frame is 30, find the value of x .

- A. 3
- B. 5
- C. 8
- D. 10

Scratch Space:

Exercise 1.4.2

A school database evaluates 100 students tracking their choices of technical electives: 55 select Robotics (R) and 48 select Accounting (A). An analyst sketches a circular visualization diagram and finds that 15 students take both.

- (a) Draw a fully labeled Venn diagram detailing the explicit element count within each partitioned layout region.
- (b) Determine how many students are placed in the outer boundary region (taking neither elective).

Solution Space:

Exercise 1.4.3

A survey of 60 tourists reveals their preferred breakfast items: 35 drink Coffee (C), 28 drink Tea (T), and 12 choose both. How many individual tourists choose neither coffee nor tea?

- A. 9
- B. 12
- C. 15
- D. 23

Scratch Space:

Exercise 1.4.4

Owen inspects a group of 45 students regarding their leisure devices: 27 own an electronic tablet, 23 own a laptop console, and 5 own neither device.

- (a) Construct a Venn diagram with a central variable placeholder to represent the double-counted core.
- (b) Calculate the precise number of students who own *both* a tablet and a laptop console.

Solution Space:

Exercise 1.4.5

A collection contains 50 total components. Set X encompasses $2k + 10$ components, and Set Y encompasses $3k$ components. If their overlapping section contains k items and no items sit outside the circles, express the universal boundary equation to find k .

- A. $4k + 10 = 50$
- B. $5k + 10 = 50$
- C. $6k = 50$
- D. $k = 50$

Scratch Space:

Exercise 1.4.6

An analytical tracking frame models a network with n active channels. Sub-segment A occupies $2n - 15$ paths, sub-segment B covers $n + 12$ paths, and their mutual intersection cross-link logs exactly 8 paths. There are 10 paths outside the circles.

- (a) Sketch a Venn diagram mapping these expressions directly into localized cells.
- (b) Solve for the structural value of n given that the universal frame contains 65 total paths.

Solution Space:

Exercise 1.4.7

A school recreation club has 180 total members[cite: 3]. Let circle M track male members and circle S track senior sports badge holders. It is discovered that 105 members are male, 80 members hold senior sports badges, and 40 male members hold senior sports badges. How many members are found outside both tracked circles?

- A. 35
- B. 40
- C. 65
- D. 145

Scratch Space:

Exercise 1.4.8

An exam registrar compiles registration files for 120 students in a cohort[cite: 4]. List P contains 75 candidates who passed the written component, and List Q contains 55 candidates who passed the practical track. A total of 12 candidates failed both tracks entirely.

- (a) Translate these metrics into a complete Venn diagram layout.
- (b) Calculate how many candidates successfully cleared the written component but failed the practical track.

Solution Space:

Exercise 1.4.9

A specialized math diagnostic set analyzes properties of integers within the collection $U = \{1, 2, 3, \dots, 24\}$. Circle X maps multiples of 3, Circle Y maps multiples of 4, and Circle Z maps multiples of 6. How many items reside within the intersection region shared by all three circles ($X \cap Y \cap Z$)?

- A. 1
- B. 2
- C. 4
- D. 6

Scratch Space:

Exercise 1.4.10

A multi-stream computer system evaluates structural logs across three error categories: Group A maps data-link errors, Group B maps database index mismatches, and Group C maps UI timeouts. The setup contains 90 total cases.

- (a) Given that 5 cases trigger all three errors, 12 trigger A and B , 10 trigger A and C , 8 trigger B and C , and the pure exclusive single-error region counts are all exactly 15, assemble a complete 3-circle Venn diagram.
- (b) Deduce the number of error logs that sit in the background universal region (no errors triggered).

Solution Space:

Exercise 1.4.11

A student is given data where $\text{Count}(A) = 20$, $\text{Count}(B) = 15$, and $\text{Count}(A \text{ and } B) = 22$. When drawing the Venn diagram, the student writes down a count of -2 in the shared intersection. This layout configuration indicates a logical failure because:

- A. Intersection spaces can only use variables.
- B. The overlap count cannot exceed an individual set's total count; negative physical elements are impossible.
- C. Circles must be drawn separated.
- D. The system requires three circles to resolve.

Scratch Space:

Exercise 1.4.12

An instructional designer looks at an improperly constructed question meant for a public test bank. The draft states: “In a class of 30 students, 25 take History, 22 take Geography, and 4 take neither.”

- (a) Use a Venn diagram calculation to expose the internal mathematical contradiction hidden within this problem statement.
- (b) Propose an alternative parameter set that resolves this layout failure while maintaining realistic student distributions.

Solution Space:

1.5 The Tree Diagram

“When choices occur in a specific order, time splits the future into branches. A tree diagram is a visual map of all sequential paths, growing outward from a single trunk to display every chronological outcome clearly.”

Definition: The Tree Diagram

A tree diagram is a branching geometric structure used to systematically list all possible outcomes of a sequence of multi-stage choices. Each stage of choice is represented by a set of diverging lines (*branches*), and each unique path from the starting root to a final tip (*leaf*) corresponds to a single, complete compound outcome.

Intuitive Core: We use tree diagrams when there is a clear *order of appearance* (e.g., first choice, second choice, third choice). By tracing along the branches, we can ensure that we do not skip any chronological combination. **WARNING: EACH STEP LATER IN THE PATH DEPENDS ON THE OUTCOME OF THE STEP PRECEDING IT. CAREFULLY TRACK HOW THE BRANCH OPTIONS SPREAD OUT!**

Example: Mapping Sequential Choices Chronologically

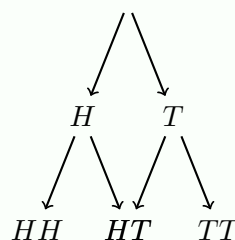
A student tosses a fair coin twice. Draw a tree diagram to list all possible structural outcomes of these two consecutive actions.

Detailed Solution Steps:

- Stage 1 (First Toss):** The coin can either land on Heads (H) or Tails (T). Draw two initial branches splitting from the starting root.
- Stage 2 (Second Toss):** From each of the first tips (H and T), draw another pair of branches splitting into H and T .
- List the final outcome paths:** Tracing each pathway from left to right yields exactly four complete sequences:

$$\text{Outcomes} = \{HH, HT, TH, TT\}$$

$$\text{Total number of paths} = 2 \times 2 = 4$$



Hint / Concept Review

Reading a tree diagram is like following roads on a map. Each completed road represents an option. This visualization prevents confusion when sequential tracking is required.

Subsection 1.5 Practice Exercises**Exercise 1.5.1**

A standard tree diagram is constructed to monitor a two-stage binary experiment. Stage 1 has 2 branches, and Stage 2 has 3 branches extending from each Stage 1 outcome. How many total unique outcome terminal leaves exist at the far right of the diagram?

- A. 5
- B. 6
- C. 8
- D. 9

Scratch Space:

Exercise 1.5.2

A game requires spinning a spinner that can land on either Red (R) or Blue (B). A player spins it twice in a row.

- (a) Construct a fully labeled tree diagram detailing the sequential paths.
- (b) List the entire sample space of final sequential outcomes based on your diagram.

Solution Space:

Exercise 1.5.3

A weather station logs daily atmospheric changes over two consecutive days. Each day can be classified as either Sunny (S) or Rainy (R). Tracing the total sequential paths over these 48 hours, how many distinct chronological outcomes are mapped?

- A. 2
- B. 4
- C. 6
- D. 8

Scratch Space:

Exercise 1.5.4

Owen plays a best-of-three tournament series. The first player to win two games wins the entire series. Each game can only end in a Win (W) or a Loss (L).

- (a) Construct a tree diagram to illustrate the tournament paths.

Note: Stop a path early if a player wins two games.

- (b) Count the total number of unique terminal paths that describe how the tournament can conclude.

Solution Space:

Exercise 1.5.5

An asset security check runs 3 sequential screenings. Each tracking stage can output either Pass (P) or Fail (F). Express the total number of final diagnostic branches in terms of a basic base-2 power calculation.

- A. $2 \times 3 = 6$
- B. $2^3 = 8$
- C. $3^2 = 9$
- D. $2 \times 2 \times 2 \times 2 = 16$

Scratch Space:

Exercise 1.5.6

A manufacturing path requires a component to go through n distinct verification checks one after the other. Each check can result in either Acceptance (A) or Rejection (R).

- (a) Formulate an expression for the total number of final outcome paths in terms of n .
- (b) If an inspector draws a full path tree and counts exactly 16 terminal leaf nodes, evaluate the parameter n .

Solution Space:

Exercise 1.5.7

A driver travels through two consecutive intersections controlled by traffic signals[cite: 2]. Each signal can display either Stop (S) or Go (G). How many complete outcome combinations can be tracked across both intersections?

- A. 2
- B. 3
- C. 4
- D. 6

Scratch Space:

Exercise 1.5.8

An examination format consists of two sequential parts: a theory paper and a subsequent practical test[cite: 1]. A student can either pass or fail each independent component.

- (a) Draft a tree diagram tracking all possible performance histories for a single candidate.
- (b) If a candidate must pass both parts to obtain the qualification[cite: 1], shade or highlight the branch pathway that satisfies this condition on your diagram, and state how many outcomes achieve this.

Solution Space:

Exercise 1.5.9

A fair four-sided die (marked with numbers 1, 2, 3, and 4) is rolled twice in a row. If a tree diagram is drawn to trace the values sequence, how many total outcome paths are generated?

- A. 8
- B. 12
- C. 16
- D. 24

Scratch Space:

Exercise 1.5.10

A multi-stage vault security protocol requires entering three sequential inputs. Input 1 can be any letter from $\{A, B\}$. Input 2 can be any digit from $\{1, 2, 3\}$. Input 3 can be any symbol from $\{\star, \#\}$.

- (a) Generate a tree diagram displaying the complete sequence of inputs.
- (b) Calculate the total number of unique combinations that can be entered into the device.

Solution Space:

Exercise 1.5.11

A student maps an experiment where they draw two marbles from a bag without replacing the first marble. The bag initially contains 1 red and 1 blue marble. The student draws two identical independent branches for Stage 2 regardless of what happened in Stage 1. This tree layout fails because:

- A. Tree diagrams cannot handle marbles.
- B. Once a marble is removed without replacement, that option disappears for the next draw, which changes the available branches.
- C. The branches should have been multiplied instead of split.
- D. Blue marbles must branch before red marbles.

Scratch Space:

Exercise 1.5.12

An instructional designer evaluates a template tree diagram meant to illustrate sequential draws without replacement. The bag contains 1 gold coin and 2 silver coins. Two coins are drawn one by one.

- (a) Explain why a path starting with drawing a gold coin cannot branch into another gold coin option in the second stage.
- (b) Sketch the correct asymmetrical tree structure to reflect this constraint accurately.

Solution Space:

1.6 Random Selection from a Finite Set: One Object

“When we pull a single item blindly from a closed box, fairness means every separate object has the exact same baseline strength to be chosen. Counting the total available objects sets our baseline universe.”

Definition: Discrete Uniform Random Selection (One Object)

Let S be a finite set containing N distinct and equal objects. If exactly one object is selected *at random*, the total number of possible outcomes is given directly by the size of the set, which is N . Every individual object has an equal opportunity of being selected.

Intuitive Core: When selecting only **ONE** object, there are no structural stages, no combinations, and no order tricks to track. The number of possibilities matches the number of objects you can physically touch. **WARNING: DO NOT APPLY MULTIPLICATION OR FACTORIALS WHEN SELECTING ONLY A SINGLE ITEM FROM A POOL!**

Example: Single Selection Count

A bag contains 8 red marbles, 5 blue marbles, and 3 green marbles. If a student randomly picks exactly one marble from the bag, find the total number of possible outcomes.

Detailed Solution Steps:

1. List the available categories in the set:

$$\text{Red} = 8, \quad \text{Blue} = 5, \quad \text{Green} = 3$$

2. Since only one object is selected, find the total size of the collection using basic arithmetic:

$$\text{Total Objects } N = 8 + 5 + 3 = 16$$

3. Therefore, the total number of possible individual selection outcomes is exactly 16.

Hint / Concept Review

Do not let the different colors confuse you! The question asks for the total possible outcomes of picking **ONE** marble. Because every single marble is distinct, your total choice pool is simply the sum of all items.

Subsection 1.6 Practice Exercises**Exercise 1.6.1**

A box contains 9 cards numbered 1, 2, 3, 4, 5, 6, 7, 8, and 9 respectively[cite: 4]. If one card is randomly drawn from the box, how many total possible outcomes are there?

- A. 1
- B. 5
- C. 9
- D. 81

Scratch Space:

Exercise 1.6.2

A standard fair six-sided die is rolled once.

- (a) Write down the complete set of all possible outcomes.
- (b) Find the total number of possible outcomes.

Solution Space:

Exercise 1.6.3

A box contains ten cards numbered 2, 3, 5, 8, 11, 11, 12, 15, 19, and 24 respectively[cite: 4]. If a card is randomly drawn from the box, how many unique options can be pulled out?

- A. 1
- B. 2
- C. 9
- D. 10

Scratch Space:

Exercise 1.6.4

Owen has a drawer containing 8 white socks, 4 yellow socks, and 2 red socks[cite: 4]. He reaches in and pulls out exactly one sock at random.

- (a) Find the total number of socks available in the drawer.
- (b) State the total number of possible individual choices Owen could make.

Solution Space:

Exercise 1.6.5

A raffle drum holds x green tickets, $2x$ blue tickets, and 5 orange tickets. Express the total number of possible outcomes for drawing a single ticket in terms of x .

- A. $3x$
- B. $2x + 5$
- C. $3x + 5$
- D. $2x^2 + 5$

Scratch Space:

Exercise 1.6.6

A container stores n defective computer chips and $n^2 - 3$ functional computer chips.

- (a) Construct a simplified algebraic expression for the total number of outcomes when selecting one chip.
- (b) If the total number of selection possibilities is 17, find the value of n .

Solution Space:

Exercise 1.6.7

There are 18 boys and 12 girls in a class[cite: 4]. If a student is randomly selected from the class to serve as the class representative, find the total number of possible individual choices[cite: 4].

- A. 12
- B. 18
- C. 30
- D. 216

Scratch Space:

Exercise 1.6.8

In a factory department, employees are split across categories: 58 are administrative officers and 142 are clerks[cite: 4]. If one employee is randomly selected for a staff interview,

- (a) Calculate the total number of unique outcomes for this selection.
- (b) Explain why the gender or age subdivisions do not alter the total outcome count as long as exactly one employee is pulled from the entire registry pool.

Solution Space:

Exercise 1.6.9

A card is drawn from a stack containing all numbers from 100 to 999 inclusive[cite: 4]. How many total possible selection outcomes exist in this finite set?

- A. 899
- B. 900
- C. 999
- D. 1000

Scratch Space:

Exercise 1.6.10

A bag contains 4 green pens, 7 blue pens, and 8 black pens.

- (a) Determine the number of possible outcomes if one pen is selected at random.
- (b) Suppose we add k purple pens to the bag, making the total number of single selection outcomes equal to 25. Find the value of k .

Solution Space:

Exercise 1.6.11

A teacher asks for the number of outcomes when picking a single item from a box containing 6 erasers and 5 pens. A student answers that there are $6 \times 5 = 30$ outcomes. This calculation is a failure because:

- A. Erasers and pens must be counted using subtraction.
- B. The student multiplied counts as if making a two-stage compound choice, but only a single object is being chosen.
- C. The items are not distinct.
- D. Additive logic requires identical sets.

Scratch Space:

Exercise 1.6.12

An analyst examines an insurance database classifying 70 aeroplanes as low-risk and 30 aeroplanes as high-risk[cite: 4]. The database picker selects exactly one aeroplane profile.

- (a) Identify the mistake in assuming that the risk categories create separate multi-stage choices when picking one single record.
- (b) Provide the correct arithmetic breakdown to justify the true universe of individual outcomes.

Solution Space:

1.7 Random Selection from a Finite Set: More than One Object with Replacement

“When we return an item to the box before picking the next one, time resets the world. Because the pool never shrinks, every stage faces the exact same universe of options, creating an exponential explosion of paths.”

Definition: Random Selection with Replacement

Let a finite set contain N distinct objects. If r objects are selected sequentially one by one, and after each selection, the chosen object is replaced back into the set before the next draw occurs, then the number of options at each stage remains exactly N . The total number of unique sequential outcomes is:

$$\text{Total Outcomes} = \underbrace{N \times N \times \cdots \times N}_{r \text{ times}} = N^r$$

Intuitive Core: Choosing *with replacement* means your selection pool is completely cloned at every step. Since the choices at Stage 1, Stage 2, $\dots$, and Stage r are all independent and identical, we apply the Law of Multiplication using identical factors, which condenses into an **exponent structure**. **WARNING: REPLACEMENT MEANS THE SAME OBJECT CAN BE CHOSEN MULTIPLE TIMES! NEVER DECREASE THE VALUE OF THE BASE POOL BETWEEN DRAWING STAGES!**

Example: Exponential Multi-Stage Selection

A box contains 6 cards numbered 1 to 6. A student draws a card randomly, records its value, and puts it back into the box. This process is repeated 3 times in total. Find the total number of unique number sequences that can be generated.

Detailed Solution Steps:

Let the total size of the set be $N = 6$.

Let the number of consecutive stages be $r = 3$.

Choices for Draw 1 = 6

Choices for Draw 2 = 6 (due to replacement)

Choices for Draw 3 = 6 (due to replacement)

Using the Law of Multiplication:

$$\text{Total Sequences} = 6 \times 6 \times 6 = 6^3 = 216$$

Hint / Concept Review

Identify the keyword *with replacement*. This tells you that the pool resets to 6 cards every single time.

Instead of writing out long strings of multiplication, we use the exponent format N^r , where the base N is the pool size and the power r is the number of operations.

Subsection 1.7 Practice Exercises**Exercise 1.7.1**

Two cards are randomly drawn from a box containing 9 cards numbered 1 to 9 one by one *with replacement*[cite: 4]. How many total unique sequential pairings can be recorded?

- A. 17
- B. 18
- C. 72
- D. 81

Scratch Space:

Exercise 1.7.2

A standard fair six-sided die is tossed 3 times in a row.

- (a) Explain why a dice-rolling sequence maps mathematically to random selection with replacement.
- (b) Calculate the total number of possible sequence profiles in the sample space.

Solution Space:

Exercise 1.7.3

A bag contains 8 white socks, 4 yellow socks, and 2 red socks[cite: 4]. A boy randomly takes out a sock, records its color, and returns it to the drawer. He repeats this exact procedure a second time. How many total outcomes describe the individual pairs of specific socks he could draw?

- A. 14
- B. 28
- C. 182
- D. 196

Scratch Space:

Exercise 1.7.4

A vault key entry pad consists of 10 distinct numeric buttons (0–9). A user creates a custom sequential password passcode by clicking buttons 4 times. Duplicate clicks are allowed.

- (a) Set up the exponential expression tracking the total number of valid passcodes.
- (b) Evaluate the final count of individual passcode options.

Solution Space:

Exercise 1.7.5

A box holds x distinct items. If an analyst draws an item at random with full replacement a total of 3 consecutive times, express the size of the compound outcome workspace in terms of x .

- A. $3x$
- B. $x + 3$
- C. x^3
- D. 3^x

Scratch Space:

Exercise 1.7.6

An online survey randomly generates system tokens by selecting from a pool of n unique items with immediate replacement.

- (a) Write an algebraic formula for the total number of outcome paths generated across 2 consecutive selections.
- (b) Given that a 2-stage replacement trial yields exactly 144 unique path sequences, evaluate the pool size parameter n .

Solution Space:

Exercise 1.7.7

A computer program randomly selects a student representative from a school register containing 625 boys and 450 girls[cite: 3]. The program draws one name, records it, resets the database pool, and then draws a student name a second time. How many unique pairs of student selections can be tracked?

- A. 1075
- B. 2150
- C. 1075^2
- D. 625×450

Scratch Space:

Exercise 1.7.8

An exam testing center has 16 distinct student question banks on their computers[cite: 1]. The server creates a customized 2-day test profile for Owen by picking 1 question bank on Day 1, replacing it into the active choices list overnight, and picking 1 question bank on Day 2.

- (a) Find the total number of unique exam assignment combinations that the server could build.
- (b) TRAP WARNING: Suppose Owen mistakenly claims that because the question banks are replaced, he can calculate the options using $16 + 16 = 32$. Mathematically explain why addition fails to model sequential multi-stage pairings.

Solution Space:

Exercise 1.7.9

A fair four-sided spinner labeled 1, 2, 3, and 4 is spun 3 times in a row. How many elements are contained in the full sequence tree mapping this compound behavior?

- A. 12
- B. 64
- C. 81
- D. 256

Scratch Space:

Exercise 1.7.10

An algorithmic password tool builds security tags by picking elements from a small structural character list containing 5 units: $\{A, B, C, \dots, p\}$. The tool generates tags by pulling items from the list 3 times, replacing the item each time.

- (a) Express the size of the total validation workspace as an exponential term depending on p .
- (b) If the tool registers exactly 343 distinct structural tags, find the value of p .

Solution Space:

Exercise 1.7.11

A student is tracking sequential choices from a set of 8 colored cards. They draw 2 cards one after another with full replacement. The student calculates the total unique sequences as $8 \times 7 = 56$. This calculation is a failure because:

- A. Exponent base logic requires adding pools.
- B. The student reduced the choice pool for the second draw by 1, failing to recognize that replacement restores the full set of 8 options.
- C. Cards are non-numeric elements.
- D. The numbers should be cubed.

Scratch Space:

Exercise 1.7.12

An analyst models options for generating automated security tags. System A requires 3 characters chosen from a pool of 6 items with replacement. System B requires 2 characters chosen from a pool of 9 items with replacement.

- (a) Use exponential counters to calculate the exact size of the total unique outcome profiles for both systems.
- (b) Compare these systems to determine which one creates a larger configuration landscape for security tracking.

Solution Space:

1.8 Random Selection from a Finite Set: Without Replacement (At Most Three Objects)

“When we keep an object after pulling it out, the universe loses a piece of itself. Because the pool shrinks with every step, each consecutive choice faces a smaller pool, causing our counting numbers to tick downward.”

Definition: Random Selection Without Replacement

Let a finite set contain N distinct objects. If more than one object is drawn sequentially one after another, and each chosen object is kept outside the pool (not replaced) before the next stage begins, the number of available choices decreases by exactly 1 at each subsequent step. The total number of unique sequential outcomes for drawing 2 or 3 objects is computed using a descending product:

$$\text{Total Outcomes (2 Objects)} = N \times (N - 1)$$

$$\text{Total Outcomes (3 Objects)} = N \times (N - 1) \times (N - 2)$$

Intuitive Core: Choosing *without replacement* breaks the identical symmetry of our options. Because an item cannot be picked twice, Stage 1 has N options, Stage 2 has $(N - 1)$ options remaining, and Stage 3 has $(N - 2)$ options left. We chain these changing pools together using the Law of Multiplication. **TRAP WARNING: THE BASE SIZE MUST DECREASE BY EXACTLY ONE AT EACH STAGE! NEVER USE EXPONENTS (N^n) WHEN REPLACEMENT IS PROHIBITED!**

Example: Descending Multi-Stage Selection

A bag contains 7 distinct colored tokens. A student draws 3 tokens sequentially one by one without replacing any back into the bag. Find the total number of unique sequential outcome paths that can be tracked.

Detailed Solution Steps:

Let the initial pool size be $N = 7$.

Choices for Draw 1 = 7

Choices for Draw 2 = $7 - 1 = 6$

Choices for Draw 3 = $6 - 1 = 5$

Using the Law of Multiplication across these changing stages:

$$\text{Total Sequences} = 7 \times 6 \times 5$$

$$\text{Total Sequences} = 42 \times 5 = 210$$

Hint / Concept Review

Look for phrases like *without replacement* or *one after another and kept*.

Imagine the physical box shrinking. Since the student removes a token, only 6 tokens remain for the second draw, and then only 5 remain for the third draw. Chaining these choices gives $7 \times 6 \times 5 = 210$.

Subsection 1.8 Practice Exercises**Exercise 1.8.1**

Two cards are randomly drawn one by one without replacement from a box containing 9 cards numbered 1 to 9[cite: 4]. How many total unique sequential pairings can be recorded in the observation log?

- A. 17
- B. 72
- C. 81
- D. 90

Scratch Space:

Exercise 1.8.2

A raffle box contains 12 unique tickets. An administrator draws 3 winning tickets sequentially, one after another, without replacement.

- (a) Set up the descending product expression used to determine the total number of unique sequences for the three winning drawings.
- (b) Calculate the final number of individual drawing sequences.

Solution Space:

Exercise 1.8.3

A drawer contains 8 white socks, 4 yellow socks, and 2 red socks[cite: 4]. Owen randomly takes out 2 socks from the drawer one by one without replacement[cite: 4]. How many total unique outcome sequences of individual socks can be drawn from the pool?

- A. 28
- B. 182
- C. 196
- D. 240

Scratch Space:

Exercise 1.8.4

A running competition has 8 runners participating in a final sprint race. The tracking system registers the individual sequences for the Gold, Silver, and Bronze medal positions.

- (a) Construct a sequential three-stage multiplication map showing how the choice pool drops at each medal assignment.
- (b) Find the total number of distinct medal outcome sequences possible.

Solution Space:

Exercise 1.8.5

A collection contains x distinct items. If an analyst draws an item at random without replacement a total of 3 consecutive times, express the size of the chronological sequence workspace in terms of a factored algebraic product.

- A. x^3
- B. $3x(x - 1)$
- C. $x(x - 1)(x - 2)$
- D. $x(x - 1)^2$

Scratch Space:

Exercise 1.8.6

A secure testing server assigns online tasks using a sequential pool selection format. The server draws items from an engineering test directory containing n available configurations without replacement.

- (a) Find a simplified algebraic expression for the total number of outcome sequences generated across 2 consecutive selections.
- (b) Given that this 2-stage sequential trial without replacement produces exactly 110 unique outcome tracks, determine the parameter n .

Solution Space:

Exercise 1.8.7

A student tracking database contains records for 21 boys and 11 girls[cite: 4]. The system randomly selects 2 individual profiles one after another without replacement to assign to consecutive interview panels. How many unique sequence outcomes can be produced?

- A. 64
- B. 992
- C. 1024
- D. 1056

Scratch Space:

Exercise 1.8.8

A multi-department corporation nominates representatives to a special advisory task force[cite: 4]. To fill the executive leadership roles of Chairperson and Vice-Chairperson, 16 candidate representatives are put into a ballot box[cite: 4]. The selections are drawn one after another.

- (a) Calculate the total number of unique pairing sequences possible for these leadership roles.
- (b) TRAP WARNING: Suppose a junior clerk suggests that because both positions are drawn from the same pool, the options can be calculated using $16^2 = 256$. Explain why this exponential calculation fails for selection without replacement.

Solution Space:

Exercise 1.8.9

A lock box holds ten cards numbered 2, 3, 5, 8, 11, 11, 12, 15, 19, and 24[cite: 4]. If a player draws 2 cards one by one from the box without replacement, how many total unique numeric index sequence pathways can occur?

- A. 20
- B. 45
- C. 90
- D. 100

Scratch Space:

Exercise 1.8.10

An experimental setup requires choosing sub-samples from a collection of items. The system pulls items sequentially without replacement from a base set of size $p + 1$. The process runs for exactly 3 consecutive stages.

- (a) Construct the algebraic product sequence representing the size of the total outcome universe.
- (b) Given that the total sequence count evaluates to 120, find the value of p .

Solution Space:

Exercise 1.8.11

A student needs to pick a captain and a co-captain from a team of 7 players. They calculate the total combinations by writing down $7 \times 7 = 49$. This calculation is a failure because:

- A. Chaining unique roles requires adding numbers.
- B. One person cannot occupy both distinct roles simultaneously; the pool must shrink for the second selection stage.
- C. Team management requires using division.
- D. The numbers must be multiplied by three.

Scratch Space:

Exercise 1.8.12

An auditor reviews data logging paths from a finite batch system. Batch X contains 6 items, and sequential draws are done without replacement.

- (a) Analyze the difference in the number of generated sequence outcomes when running a 2-stage draw versus a 3-stage draw.
- (b) Explicitly show the products for both cases to demonstrate how each extra stage continues to decrease the available pool size.

Solution Space:

1.9 Permutation of a Set: Factorial Only

“When we arrange every single object in a collection into a complete line, no item is left behind. A full arrangement is simply a cascading countdown of choices, multiplying until our pool drops down to the very last element.”

Definition: The Factorial Notation

The total number of unique ways to arrange an entire set of n distinct objects into a specific sequence or linear order is denoted by $n!$ (read as “ n factorial”) and is calculated as the descending product of all positive integers from n down to 1:

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1$$

By definition, we anchor the empty arrangement base case as $0! = 1$.

Intuitive Core: A full arrangement is just an extension of selection without replacement where we draw **EVERY** item until the box is empty. For the first slot in line, you have n choices; for the second, $(n-1)$; and this countdown continues until only 1 choice remains for the final slot. **WARNING: FACTORIALS APPLIES ONLY WHEN YOU ARE ARRANGING ALL OBJECTS IN THE SET! DO NOT STOP THE DOWNWARD CASCADE EARLY IF EVERY ITEM MUST BE IN THE LINE!**

Example: Arranging a Complete Collection

There are 5 distinct math textbooks sitting on a shelf. In how many unique linear ways can a tutor arrange all 5 books from left to right?

Detailed Solution Steps:

Let the total number of distinct books to arrange be $n = 5$.

Choices for Slot 1 = 5

Choices for Slot 2 = 4

Choices for Slot 3 = 3

Choices for Slot 4 = 2

Choices for Slot 5 = 1

Using the factorial formula: $5! = 5 \times 4 \times 3 \times 2 \times 1$

$5! = 120$ ways

Hint / Concept Review

Arranging things in a row means that the order matters.

Because you are sorting all 5 items, the problem maps directly to $5!$. Basic arithmetic confirms that $5 \times 4 = 20$, then $20 \times 3 = 60$, $60 \times 2 = 120$, and $120 \times 1 = 120$.

Subsection 1.9 Practice Exercises**Exercise 1.9.1**

An eight-digit security code is formed by a complete permutation of the digits 2, 3, 4, 5, 6, 7, 8, and 9 where no digit can be repeated[cite: 4]. How many different eight-digit codes can be generated?

- A. 8
- B. 64
- C. 5040
- D. 40320

Scratch Space:

Exercise 1.9.2

Four students need to stand in a single straight queue at a canteen counter.

- (a) Write out the descending product representing the full structural arrangement permutations of this line.
- (b) Evaluate the total number of unique queue combinations.

Solution Space:

Exercise 1.9.3

A teacher has 6 unique award badges to distribute to 6 selected students. If each student must receive exactly one unique badge, how many different ways can the badges be assigned?

- A. 36
- B. 720
- C. 5040
- D. 46656

Scratch Space:

Exercise 1.9.4

Owen is hanging 7 different paintings in a single horizontal row along a main hallway.

- (a) Set up the factorial expression tracking the complete layout configuration space.
- (b) Find the total number of distinct visual arrangements possible.

Solution Space:

Exercise 1.9.5

A set consists of $x + 1$ distinct elements. Express the total number of unique ways to arrange every single element of this set into a linear row in terms of x .

- A. $x!$
- B. $(x + 1) \times x$
- C. $(x + 1)!$
- D. $(x!) + 1$

Scratch Space:

Exercise 1.9.6

An automated test registry organizes a stack of student test portfolios for filing. The stack contains n files.

- (a) Write down an algebraic factorial expression for the total number of ways to order the stack.
- (b) Given that the total number of valid ordering permutations evaluates to 24, determine the value of n .

Solution Space:

Exercise 1.9.7

A group of 5 distinct numbers are to be arranged to form a five-digit passcode. If no numbers can be repeated, how many total passcode arrangements can be constructed?

- A. 25
- B. 120
- C. 600
- D. 3125

Scratch Space:

Exercise 1.9.8

An individual photo session requires arranging a complete group of students in a single line. The group consists of 3 boys and 3 girls.

- (a) Compute the total number of unique arrangements if there are no structural restrictions on who sits where.
- (b) TRAP WARNING: Suppose a student claims that since there are two gender categories, the total linear paths can be found by evaluating $3! \times 2!$. Explain clearly why this logic fails to capture the true permutation space of all 6 distinct individuals.

Solution Space:

Exercise 1.9.9

A mathematical puzzle card set consists of 4 unique tracking symbols. If an algorithm is programmed to output every single possible full linear permutation of these cards one after another, how many unique outputs will the algorithm display?

- A. 4
- B. 16
- C. 24
- D. 256

Scratch Space:

Exercise 1.9.10

A custom code block sorts letters linearly. The base character set contains a sequence of k letters.

- (a) State the simplified expression representing the configuration boundaries of a full stack permutation.
- (b) If the total arrangement space size expands to 720 options, find the value of k .

Solution Space:

Exercise 1.9.11

A student wants to find the number of ways to arrange 3 identical red ball markers in a row. They apply the standard factorial rule and state the answer is $3! = 6$. This calculation is a failure because:

- A. Factorials cannot be computed on small sets.
- B. Swap configurations of identical items do not produce visually distinct new sequences; the factorial rule assumes all items are completely unique.
- C. The items must be added.
- D. Identical items require exponential base terms.

Scratch Space:

Exercise 1.9.12

An analyst examines an error in an automated scheduling database. The system calculates the number of ways to line up 4 distinct tasks as $4 \times 4 = 16$.

- (a) Discuss the structural flaw in using exponential arithmetic for an arrangement problem where tasks cannot run simultaneously or be duplicated in the timeline.
- (b) Provide the correct factorial configuration step to demonstrate how the available selection pool drops as each slot in the sequence is populated.

Solution Space:

1.10 Permutation from a Set: nPr

“When we only need to arrange a small group chosen out of a much larger crowd, the countdown cascade stops early. We don’t multiply all the way down to one; we drop the numbers only for the specific slots we need to fill.”

Theorem: The Permutation Formula (nPr)

The total number of unique ways to select and arrange exactly r items in a specific linear order from a pool of n distinct objects ($0 \leq r \leq n$) is denoted by nPr or $\frac{P_r^n}{}$ and is given by a descending product containing exactly r factors:

$$nPr = \underbrace{n \times (n-1) \times (n-2) \times \cdots \times (n-r+1)}_{r \text{ factors}} = \frac{n!}{(n-r)!}$$

Intuitive Core: nPr is just selection without replacement where we stop after filling exactly r slots. If you have 10 items but only fill 3 slots, you have 10 options for the first slot, 9 for the second, and 8 for the third. Chaining them gives $10 \times 9 \times 8$, which is exactly $10P3$. **WARNING: ORDER MATTERS INDEPENDENTLY! SWAPPING THE POSITION OF TWO CHOSEN ITEMS CREATES A COMPLETELY NEW AND SEPARATE ARRANGEMENT PROFILE!**

Example: Partial Arrangement out of a Pool

A committee consists of 7 distinct members. In how many ways can a tutor choose and arrange a Chairperson, a Secretary, and a Treasurer from this pool?

Detailed Solution Steps:

Let the total number of distinct members be $n = 7$.

Let the number of specific roles to fill be $r = 3$.

Choices for Chairperson = 7

Choices for Secretary = 6

Choices for Treasurer = 5

Using the permutation formula:

$$7P3 = 7 \times 6 \times 5 = \frac{7!}{(7-3)!} = 210 \text{ways}$$

Hint / Concept Review

Because each position holds a distinct, named title, the **order of assignment matters**.

Instead of computing the full factorial $7!$, we execute a descending product with exactly 3 factors since only 3 unique slots are filled.

Subsection 1.10 Practice Exercises**Exercise 1.10.1**

An eight-digit phone number is formed by a permutation of the distinct digits 2, 3, 4, 5, 6, 7, 8, and 9[cite: 4]. If we want to find the number of ways to create a short 3-digit regional extension code using these digits without repetition, which expression gives the total options?

- A. 8^3
- B. $8P3$
- C. 3^8
- D. $8!$

Scratch Space:

Exercise 1.10.2

A classroom has 15 distinct chairs. There are 3 students who need to be seated, each occupying exactly one chair.

- (a) Formulate the specific nPr term required to count the total unique seating arrangements.
- (b) Compute the exact number of seating permutations.

Solution Space:

Exercise 1.10.3

A photography team wants to select a captain and a vice-captain from a class committee containing 5 teachers from School A and 4 teachers from School B [cite: 2]. How many unique ways can these two distinct leadership roles be assigned?

- A. 9
- B. 18
- C. 72
- D. 81

Scratch Space:

Exercise 1.10.4

Owen operates a local boutique that tracks staff shifts. There are 10 salesgirls available on a particular day. The morning shift requires 1 supervisor, 1 cashier, and 1 floor monitor. Each role must be filled by a unique person.

- (a) Write out the descending factor calculation for assigning these positions.
- (b) Evaluate the total number of distinct scheduling permutations.

Solution Space:

Exercise 1.10.5

A collection contains x distinct items. An analyst needs to choose and arrange exactly 2 items into a labeled sequence. Express the total configuration possibilities as a polynomial in terms of x .

- A. x^2
- B. $2x$
- C. $x^2 - x$
- D. $x!$

Scratch Space:

Exercise 1.10.6

An engineering directory handles database entries through index slots. A register blocks out 3 distinct priority queues from a pool of n server paths.

- (a) Construct a simplified algebraic product tracking the total configuration pathways available.
- (b) If the total configuration size equals 120 unique sequences, determine the integer value of n .

Solution Space:

Exercise 1.10.7

There are 18 boys and 12 girls in a class cohort[cite: 4]. To run an inter-school debate, the teacher must choose 1 prime minister speaker, 1 deputy speaker, and 1 government whip speaker from the pool of boys only. Find the number of unique leadership line-ups possible.

- A. $30P3$
- B. $18P3$
- C. 18^3
- D. $12P3$

Scratch Space:

Exercise 1.10.8

An art curator wants to mount a small decorative showcase along a wall. There are 12 unique historical paintings available in storage. The showcase line has exactly 4 distinct hanging slots.

- (a) Calculate the total number of unique visual permutations that can be displayed.
- (b) TRAP WARNING: A trainee assumes that because we are only hanging 4 out of 12 paintings, the total calculation is simply $12 \times 4 = 48$. Explain why this completely fails to capture the multi-stage cascade of choice without replacement.

Solution Space:

Exercise 1.10.9

A password cracking algorithm logs code flags using distinct elements. If a flag is formed by taking 4 distinct units from the symbol set $\{A, B, C, D, E, F, G\}$, how many unique linear sequence arrangements can be logged?

- A. 28
- B. 35
- C. 840
- D. 2401

Scratch Space:

Exercise 1.10.10

A tracking system builds operational paths. The tool maps sequential operations by picking 2 distinct keys from an asset array containing $k + 1$ tracking elements.

- (a) Express the size of the permutation output field as an algebraic formula in terms of k .
- (b) Given that the configuration database logs exactly 56 unique sequences, evaluate the parameter k .

Solution Space:

Exercise 1.10.11

A teacher wants to choose a team captain and an equipment monitor from a group of 6 students. A student writes down the option count as $6C2 = 15$. This calculation is a failure because:

- A. $6C2$ treats the roles as identical, ignoring the fact that swapping who is captain and who is monitor creates a different arrangement.
- B. Permutations require full exponents.
- C. The items should be added instead.
- D. Captains cannot be permuted with monitors.

Scratch Space:

Exercise 1.10.12

An analyst runs an optimization report on scheduling tasks. The pool consists of 6 distinct tasks.

- (a) Evaluate and compare the total outcome profiles when arranging exactly 2 tasks versus arranging exactly 5 tasks.
- (b) Write out the descending products for both scenarios to show how stopping the countdown early changes the structural size of the tracking workspace.

Solution Space:

1.11 Combination from a Set: C_r^n

“When we pick a team where everyone plays the exact same role, individual order loses all meaning. A combination is simply a collection of items where who gets picked first or last does not matter.”

Theorem: The Combination Formula (C_r^n)

The total number of unique ways to select a group of r items from a pool of n distinct objects ($0 \leq r \leq n$), without any regard to the order of selection, is denoted by C_r^n or $\binom{n}{r}$ and is given by:

$$C_r^n = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$$

Intuitive Core: When order is stripped away, we use the word “CHOOSE”. If we select 3 items out of 10, computing $P_3^{10} = 10 \times 9 \times 8 = 720$ counts every group $3! = 6$ times because it treats arrangements like $\{A, B, C\}$ and $\{B, C, A\}$ as different. Dividing by $3!$ removes this layer, giving $\frac{720}{6} = 120$ unique teams. **WARNING: REARRANGING THE SELECTED ITEMS DOES NOT CREATE A NEW COMBINATION PROFILE! IF ORDER MATTERS, SWITCH BACK TO P_r^n IMMEDIATELY!**

Example: Selecting an Unordered Team

A tutor wants to select a group of 4 students from a class of 6 outstanding candidates to join an arithmetic quiz team. How many unique team combinations can be formed?

Detailed Solution Steps:

Let the total number of distinct candidates be $n = 6$.

Let the size of the team to choose be $r = 4$.

First, compute the ordered permutations:

$$P_4^6 = 6 \times 5 \times 4 \times 3 = 360$$

Next, calculate the internal duplication order variations:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Divide the ordered count by the duplication factor:

$$C_4^6 = \frac{360}{24} = 15 \text{ combinations}$$

Hint / Concept Review

Because every selected student holds the same role within the quiz team, swapping their selection order yields the identical team.

Therefore, order does not matter, which means we must apply C_r^n to safely divide out the redundant layout clones.

Subsection 1.11 Practice Exercises**Exercise 1.11.1**

There are 21 boys and 11 girls in a class[cite: 4]. If a teacher wants to select 5 students to form a special project group without any role distinctions, which mathematical term gives the total unique group combinations?

- A. P_5^{32}
- B. C_5^{32}
- C. 32^5
- D. $21! \times 11!$

Scratch Space:

Exercise 1.11.2

A committee consists of 5 teachers from School A and 4 teachers from School B [cite: 4]. A team of 4 teachers needs to be randomly selected from the entire committee[cite: 4].

- (a) Formulate the exact expression needed to calculate the total choices of choosing this group of 4.
- (b) Calculate the absolute number of unique unordered combinations possible.

Solution Space:

Exercise 1.11.3

From a task group containing 16 nominated members, a sub-committee of 4 members is randomly selected[cite: 4]. How many unique combinations of members can be formed?

- A. 64
- B. 1820
- C. 43680
- D. 65536

Scratch Space:

Exercise 1.11.4

Owen reviews a math pool box containing 11 blue cups and 5 white cups[cite: 4]. He wants to draw a batch of 6 cups from the box at the same time[cite: 4].

- (a) State whether this layout problem requires permutations or combinations, and justify why.
- (b) Evaluate the total number of distinct cup combination profiles he could pull out.

Solution Space:

Exercise 1.11.5

A master set contains $x + 2$ distinct elements. An analyst needs to select an unordered group of 2 elements from the set. Express the total combination space as a simplified polynomial in terms of x .

- A. $x^2 + 2x$
- B. $\frac{x^2+3x+2}{2}$
- C. $x^2 + 3x + 2$
- D. $\frac{x^2+x}{2}$

Scratch Space:

Exercise 1.11.6

A secure networking server draws a batch of 2 distinct tracking files from a central database bank containing n available entries.

- (a) Construct a simplified algebraic expression for the total combination possibilities.
- (b) If the distinct count of unordered file groupings evaluates to 45, find the value of n .

Solution Space:

Exercise 1.11.7

A box contains 5 red bowls, 6 yellow bowls, and 3 white bowls[cite: 4]. If 4 bowls are randomly drawn from the box at the same time[cite: 4], how many total unique group choices are available?

- A. 1001
- B. 24024
- C. 38416
- D. 576480

Scratch Space:

Exercise 1.11.8

A tournament director organizes a sports draft pool. There are 18 boys and 12 girls available in the class cohort[cite: 4]. The director needs to form a temporary committee containing exactly 4 members[cite: 4].

- (a) Find the number of unique combinations if the committee must consist of boys only[cite: 4].
- (b) Compute the combinations if the committee must contain 2 boys and 2 girls.

Solution Space:

Exercise 1.11.9

An inventory tracker logs component batches. If 5 components are randomly drawn at the same time from a storage bin holding 4 green pens, 7 blue pens, and 8 black pens[cite: 4], how many total unique combinations can be logged in the system?

- A. 95
- B. 3876
- C. 11628
- D. 139536

Scratch Space:

Exercise 1.11.10

A specialized software script forms testing arrays by choosing a sub-group of 2 elements from a resource list containing k active items.

- (a) State the combination formula matching this asset tracking challenge in terms of k .
- (b) Given that the configuration database logs exactly 28 unique combinations, evaluate the parameter k .

Solution Space:

Exercise 1.11.11

A student wants to select a committee of 3 members from a pool of 8 students. They write down the option count as $P_3^8 = 336$. This calculation is a failure because:

- A. Permutations should only be used if pool sizes grow.
- B. The student forgot that the committee roles are identical, meaning order is irrelevant and the duplication factor $3!$ must be divided out.
- C. The options must be added together.
- D. Exponent counters are required instead.

Scratch Space:

Exercise 1.11.12

An analyst evaluates options for generating automated audit profiles from a pool of 7 distinct accounts.

- (a) Evaluate the difference in unique configuration paths when generating a 3-account summary panel versus a 4-account summary panel.
- (b) Mathematically demonstrate why C_3^7 and C_4^7 yield the exact same numerical result using basic arithmetic.

Solution Space:

1.12 The Law of Division

“When items are pulled in an ordered sequence, they form an explicit layout track. Division is our mathematical filter to completely compress away this tracking layer when individual slots play identical roles.”

Theorem: The Law of Division (Symmetry Reduction)

If a structural configuration pool of size W is generated by a multi-stage process that accidentally tracks internal ordering positions among r identical elements, the pool counts every unique combination exactly $r!$ times. The true total number of distinct structural groups is obtained by dividing out the redundancy factor:

$$\text{True Unique Configurations} = \frac{W}{r!}$$

Intuitive Core: Division is the arithmetic tool we use to strip away *unwanted labels*. If an algorithm generates sequences like (A, B) and (B, A) , but our project criteria treats both tracks as the same identical pair, we divide our raw count by $2! = 2$ to collapse these duplicate records. **WARNING: ONLY DIVIDE BY THE FACTORIAL CORRESPONDING TO THE NUMBER OF SLOTS THAT SHARE AN IDENTICAL STATUS! NEVER MIX PERMUTATION SYMMETRIES!**

Example: Collapsing Ordered Tracks into Unordered Groups

An office assistant lists all possible two-person cleaning teams chosen from a small department. The database registry lists 30 ordered pathways. Since the roles are identical, use division to discover the true count of distinct teams.

Detailed Solution Steps:

Let the raw count of ordered tracking states be $W = 30$.

Let the number of identical slots in the team be $r = 2$.

Calculate the duplication factor created by internal arrangements:

$$r! = 2! = 2 \times 1 = 2$$

Apply the Law of Division to filter out the redundant orderings:

$$\text{True Unique Teams} = \frac{30}{2!} = \frac{30}{2} = 15$$

Hint / Concept Review

The database tracks pairings like (Alice, Bob) and (Bob, Alice) as two separate lines.

Because a cleaning team has no named roles, these two lines describe the same physical group. Dividing by 2 cleans up the data.

Subsection 1.12 Practice Exercises**Exercise 1.12.1**

A software script logs 120 unique ways to assign three specific student awards to a group. If the teacher changes the rules so that the three awards are completely identical, the raw count must be divided by which factor to find the unique student combinations?

- A. 3
- B. 6
- C. 9
- D. 24

Scratch Space:

Exercise 1.12.2

A math pool puzzle determines that the total number of ordered pathways to arrange 3 books from a large collection is 504.

- (a) State the internal sorting redundancy factor generated when the books are placed in an unordered stack.
- (b) Use the Law of Division to find the total number of unique unordered card combinations.

Solution Space:

Exercise 1.12.3

An administrator uses a permutation formula to find that there are 336 ways to pick a president, vice-president, and secretary from a class[cite: 4]. If the positions are changed into an identical 3-person committee, how many unique group selections exist?

- A. 14
- B. 56
- C. 112
- D. 168

Scratch Space:

Exercise 1.12.4

Owen constructs a table of permutations representing pairs of elective courses. The system tracks 90 ordered pairings.

- (a) Explain why dividing by $2!$ accurately collapses this database list when the picking order of courses is disregarded.
- (b) Calculate the total number of unique unordered course combinations.

Solution Space:

Exercise 1.12.5

A system logs M ordered pathways for selecting a group of k items. If the ordering labels among these chosen items are completely stripped away, express the number of true unique combinations in terms of M and k .

- A. $M - k!$
- B. $\frac{M}{k}$
- C. $\frac{M}{k!}$
- D. $M \times k!$

Scratch Space:

Exercise 1.12.6

An engineering data channel tracks sequential connections. The raw matrix records $x^2 - x$ ordered dual-connection events across n networks.

- (a) Formulate the algebraic expression for the unique un-ordered connection pairs by using division.
- (b) Given that the unique unordered pairs count evaluates to 21, find the value of x .

Solution Space:

Exercise 1.12.7

A tournament scheduler counts 132 unique ways to assign a first-place and second-place award to a class cohort[cite: 4]. If the director decides to give out 2 identical runner-up medals instead, find the number of unique student pairings.

- A. 11
- B. 22
- C. 66
- D. 130

Scratch Space:

Exercise 1.12.8

A school database logs student registration data. The system tracks 720 ways to choose a 3-member executive board where each role is named.

- (a) Calculate the total number of unique ways to select the board if all three members hold an identical co-director status.
- (b) TRAP WARNING: A data clerk argues that because there are 3 positions, the total count should be divided by 3 instead of $3!$. Explain the mathematical flaw in this argument.

Solution Space:

Exercise 1.12.9

A cipher generates 24 codes by rearranging a specific selection of characters. If 3 of the selected characters are completely identical, how many visually unique codes are actually produced by the cipher?

- A. 4
- B. 6
- C. 8
- D. 12

Scratch Space:

Exercise 1.12.10

An optimization engine logs W ordered tracks when choosing sub-samples. The tool collapses these tracks into unordered sets of size 4 by applying a division step.

- (a) Write down the exact numerical divisor needed to remove the internal sorting redundancy.
- (b) Given that the raw ordered count W equals 840, calculate the unique combinations.

Solution Space:

Exercise 1.12.11

A student wants to find the number of unique combinations for a 2-person project team chosen from a class. They know the ordered permutations list contains 42 pairs, so they write down $42 - 2 = 40$ combinations. This calculation is a failure because:

- A. Order variations must be removed using division rather than subtraction.
- B. The student should have added the numbers.
- C. Exponent configurations are needed.
- D. Teams can only be processed using products.

Scratch Space:

Exercise 1.12.12

An analyst evaluates registration arrays from an online system. The server logs 60 ordered paths for assigning 3 student profiles to identical study cubicles.

- (a) Discuss the critical importance of using a factorial divisor ($3!$) instead of a scalar divisor (3) when processing identical groupings.
- (b) Perform the full division step to establish the true number of unique assignment combinations.

Solution Space:

1.13 The Slot Method

“When items face restrictions or specific layout rules, we cannot just look at the whole set at once. Drawing individual physical slots lets us solve the problem piece by piece, filling the most restricted spaces first.”

Definition: The Slot Method

The slot method is a structural counting technique where each position in a linear arrangement is represented by a blank space or box ($\square$). Counting proceeds chronologically by analyzing the exact number of valid objects available to fill each slot, beginning with slots that have strict constraints or restrictions.

Intuitive Core: Think of this method as filling chairs in a row. If a slot has a special rule (e.g., “must be an even number” or “must be a vowel”), you must calculate that specific slot first. Once the restricted chairs are filled, you fill the remaining seats using the standard countdown pool. **WARNING: ALWAYS COMPLETE THE SLOTS WITH RESTRICTIONS FIRST! IF YOU FILL THE UNRESTRICTED SLOTS FIRST, YOU WILL UNINTENTIONALLY SHRINK THE CHOSEN POOL FOR THE RESTRICTED SLOTS!**

Example: Filling Slots with Boundary Constraints

How many unique 3-digit codes can be formed using the digits 1, 2, 4, 5, and 7 without repetition if the code must be an even number?

Detailed Solution Steps:

1. Draw three empty structural slots:

$\square \quad \square \quad \square$

2. Identify the restricted slot: The code must be even, so the last slot can only be filled by the digits 2 or 4.

Choices for Slot 3 = 2

3. Fill the remaining slots using the remaining elements from our pool:

Choices for Slot 1 = $5 - 1 = 4$

Choices for Slot 2 = $4 - 1 = 3$

4. Multiply across all slots:

Total Codes = $4 \times 3 \times 2 = 24$

Hint / Concept Review

The requirement “must be even” locks your attention onto the very last unit position.

By placing a 2 in the final choice slot first, you use up one even number. The other slots can then be filled normally with the 4 remaining distinct digits.

Subsection 1.13 Practice Exercises**Exercise 1.13.1**

A 3-digit passcode is formed by a permutation of the digits 3, 4, 5, 6, 7, and 8 without repetition. If the passcode must be an odd number, how many total code configurations can be generated?

- A. 40
- B. 60
- C. 120
- D. 216

Scratch Space:

Exercise 1.13.2

Five students, including Owen and a group of friends, stand in a straight row for a photograph.

(a) If Owen must stand exactly at the far-left position of the line, use a slot map to determine the remaining arrangement options.

(b) Calculate the total number of unique valid configurations for this photo setup.

Solution Space:

Exercise 1.13.3

How many unique 4-letter registration plates can be generated using the letters A, B, C, D, E, and F without repetition if the plate must begin with a vowel?

- A. 24
- B. 120
- C. 240
- D. 360

Scratch Space:

Exercise 1.13.4

An instructional task sheet requires sorting 6 distinct computer modules into 6 consecutive time slots on a testing machine. Modules X and Y are high-priority and must occupy the first two time slots in any order.

- (a) Diagram the layout choices using separate boxes for the prioritized and non-prioritized stages.
- (b) Find the total number of valid scheduling arrangements.

Solution Space:

Exercise 1.13.5

A 4-digit numeric index is built by arranging digits from the set $\{1, 2, 3, 5, 7, 8\}$ without repetition. If the index code must be greater than 5000, express the total choice tracks using a slot calculation.

- A. $3 \times P_3^5$
- B. $4 \times P_3^5$
- C. 3×6^3
- D. P_4^6

Scratch Space:

Exercise 1.13.6

An online security platform builds passcodes using n distinct letters. The system creates a 3-character security key. The letter 'Z' is reserved and must always sit exactly in the middle slot.

- (a) Construct a simplified algebraic expression representing the valid code configurations in terms of n .
- (b) If the total configuration size equals 42 unique keys, find the total letter pool size n .

Solution Space:

Exercise 1.13.7

A group of 4 boys and 3 girls line up in a straight queue[cite: 4]. If a girl must stand at the front of the queue and a boy must stand at the end of the queue, how many unique arrangements can be formed?

- A. 12
- B. 144
- C. 1440
- D. 5040

Scratch Space:

Exercise 1.13.8

An exam committee board arranges 7 distinct math test packages sequentially along a shipping counter. There are 4 core packages and 3 elective packages. The packing rules state that the first slot and the last slot must be occupied by core packages.

- (a) Evaluate the total number of unique shipping arrangements that satisfy the packing rule.
- (b) TRAP WARNING: A clerk argues that because there are 7 packages, the first slot has 4 options and the last has 3 options, so the middle slots are simply written as $5!$. Explain why this reasoning is completely invalid if a core package cannot be duplicated.

Solution Space:

Exercise 1.13.9

A cipher code uses a total of 5 slots. The available character set contains 8 distinct alphanumeric characters. If the code must start with the character 'A' and end with an even digit from the set $\{2, 4, 6\}$, how many unique code configurations can be logged?

- A. 360
- B. 1080
- C. 2160
- D. 6720

Scratch Space:

Exercise 1.13.10

A specialized tool builds asset tags using a sequence of 4 distinct slots. The pool consists of k distinct identification numbers. The first slot is restricted and can only accept a prime number chosen from a sub-list of 3 available primes.

- (a) State the simplified polynomial tracking the total configuration space in terms of k .
- (b) Given that the configuration space logs exactly 180 unique tags, find the value of k .

Solution Space:

Exercise 1.13.11

A student wants to find the number of 3-digit numbers formed from the digits 1, 2, 3, 4, and 5 without repetition that are divisible by 5. They fill the first slot with 5 options, the second with 4 options, and the third with 1 option, obtaining $5 \times 4 \times 1 = 20$. This calculation is a failure because:

- A. Divisibility logic requires adding sets.
- B. By filling the unrestricted first slot before the restricted last slot, the student mistakenly included numbers where the digit 5 was already placed in the first slot, leaving 0 choices for the final slot.
- C. Slots must use exponential base terms.
- D. The numbers must be divided by three.

Scratch Space:

Exercise 1.13.12

An analyst evaluates scheduling patterns for a series of 5 unique network checks. Checks A and B are high-risk and cannot be placed in the final slot under any circumstances.

- (a) Discuss how to prioritize your slot-filling sequence to satisfy this safety constraint without double-counting.
- (b) Construct a box-by-box breakdown showing the correct countdown product for all 5 operational positions.

Solution Space:

1.14 The Insertion Method

“When items must be kept completely separated, trying to force them into slots can lead to messy calculations. It is much easier to arrange the unrestricted items first, creating a row of empty spaces where the restricted items can be safely tucked away.”

Definition: The Insertion Method (Gap Method)

The insertion method is an arrangement technique used when certain objects must be *completely separated* from one another. We first arrange all the unrestricted elements in a linear row. This baseline row creates a series of empty spaces or spaces between items (*gaps*), including the two outer ends. We then select and arrange the restricted objects directly into these empty spaces.

Intuitive Core: If 4 boys must be completely separated from each other when standing in a line with 5 girls, do not try to guess their positions. Line up the 5 girls first: ★ G ★ G ★ G ★ G ★ G ★. This row creates exactly $5 + 1 = 6$ potential gaps (★). Since no two boys can stand in the same gap, choose and arrange the boys into these available spaces. **WARNING: NEVER PLACE MORE THAN ONE RESTRICTED OBJECT INTO THE SAME EMPTY GAP, OR ELSE THEY WILL STAND ADJACENT TO EACH OTHER AND VIOLATE THE SEPARATION RULE!**

Example: Separating Elements via Gaps

There are 4 distinct history books and 3 distinct science books to be arranged in a straight line on a shelf. If the 3 science books must be completely separated from each other, find the total number of unique arrangements.

Detailed Solution Steps:

1. Arrange the unrestricted history books first:

$$\text{Arrangements} = 4! = 24$$

2. Identify the number of empty gaps created:

__ H __ H __ H __ H __

4 history books create exactly $4 + 1 = 5$ gaps.

3. Choose and arrange the 3 science books into these 5 available gaps:

$$\text{Gap Arrangements} = P_3^5 = 5 \times 4 \times 3 = 60$$

4. Multiply the two stages together:

$$\text{Total Arrangements} = 24 \times 60 = 1440$$

Hint / Concept Review

Always remember that a row of n items creates exactly $n + 1$ gaps, counting the two outer ends.

By locking the history books down first, we create solid dividers. Dropping the science books into separate gaps ensures they can never touch each other.

Subsection 1.14 Practice Exercises**Exercise 1.14.1**

There are 5 distinct red cards and 2 distinct blue cards to be lined up in a row. If the 2 blue cards must be completely separated from each other, how many unique linear arrangements can be formed?

- A. 120
- B. 240
- C. 720
- D. 5040

Scratch Space:

Exercise 1.14.2

A student committee consists of 6 girls and 3 boys who are lining up in a straight queue for a school assembly. The 3 boys must be completely separated from one another.

- (a) State the number of empty spaces created after the 6 girls are arranged in the line.
- (b) Calculate the total number of unique ways to arrange the entire committee.

Solution Space:

Exercise 1.14.3

Owen is scheduling a sequence of 5 core math lectures and 3 elective workshops. If no two elective workshops can be placed next to each other, how many unique schedules can be constructed?

- A. 720
- B. 14400
- C. 40320
- D. 241920

Scratch Space:

Exercise 1.14.4

A set of letters containing 4 distinct vowels and 4 distinct consonants is to be arranged in a horizontal line. The 4 vowels must be completely separated from each other.

- (a) Formulate the permutation products for both the baseline stage and the gap insertion stage using the correct P_r^n notation.
- (b) Compute the total number of valid linear letter permutations.

Solution Space:

Exercise 1.14.5

There are x identical white blocks and 2 distinct black blocks. If the white blocks are placed in a fixed line and the 2 distinct black blocks must be completely separated by the white blocks, express the number of ways to insert the black blocks into the gaps.

- A. P_2^x
- B. P_2^{x+1}
- C. C_2^{x+1}
- D. $(x+1)^2$

Scratch Space:

Exercise 1.14.6

An automated script handles server checks. It arranges n standard data entries linearly and inserts 3 high-security logs such that no two high-security logs are adjacent.

- (a) Construct a simplified algebraic product for the total ways to arrange the entire sequence, assuming all data entries are distinct and all security logs are distinct.
- (b) If the number of data entries is $n = 4$, evaluate the total number of valid linear configurations.

Solution Space:

Exercise 1.14.7

A group of 5 teachers from School A and 3 teachers from School B stand in a row for a group photo[cite: 2]. If the teachers from School B must be completely separated from each other, find the number of unique photography lineups that can be made.

- A. 720
- B. 14400
- C. 17280
- D. 40320

Scratch Space:

Exercise 1.14.8

An examination hall coordinator must seat a row of 8 candidates[cite: 4]. There are 5 candidates taking a Compulsory Math paper and 3 candidates taking an Elective Math module. The 3 Elective Math candidates must be completely separated from each other.

- (a) Calculate the total number of unique seating arrangements for the 8 candidates.
- (b) TRAP WARNING: A junior supervisor suggests that since the 3 elective candidates must be separated, we can just calculate the total arrangements as $8! - 6! \times 3!$ by subtracting the ways they all sit together. Explain why this subtraction logic fails to enforce complete separation.

Solution Space:

Exercise 1.14.9

A rack has space for 9 distinct decorative pins. There are 6 green pins and 3 yellow pins. If the rack must be filled such that all yellow pins are completely separated, how many unique pin layouts are possible?

- A. 5040
- B. 30240
- C. 151200
- D. 362880

Scratch Space:

Exercise 1.14.10

A custom network code block sequences server assets. The script lines up k identical standard assets and inserts 2 distinct priority tokens into the available gaps so that they are completely separated.

- (a) Express the number of unique tag sequences in terms of k .
- (b) Given that the configuration script produces exactly 56 unique sequences, evaluate the asset parameter k .

Solution Space:

Exercise 1.14.11

A teacher asks for the number of ways to arrange 4 distinct physics books and 2 distinct chemistry books in a row such that the chemistry books are completely separated. A student answers $6! - 2! \times 5! = 480$. This calculation is a failure because:

- A. Books cannot be handled using subtraction rules.
- B. The student subtracted the ways where *all* restricted items are grouped together, which only removes the case where both chemistry books are adjacent. While that works for exactly 2 items, the logic breaks down entirely when separating 3 or more items because it forgets to track cases where only some items are adjacent.
- C. The gap count must use exponents.
- D. Separating items requires division instead of products.

Scratch Space:

Exercise 1.14.12

An instructional analyst evaluates a scheduling error in a calendar app. The app tries to place 5 standard work tasks and 3 rest breaks in a row such that no two rest breaks occur consecutively.

- (a) Analyze how the insertion method avoids creating empty adjacent slots when managing constraints.
- (b) Provide the correct factor product map to prove the true number of unique calendar configurations.

Solution Space:

1.15 Round Table Symmetry

“When paths wrap around to meet themselves, the ideas of ‘beginning’ and ‘end’ vanish. Circular logic forces us to anchor one point in space, breaking the endless wheel of rotations so that standard arithmetic can make sense of the crowd.”

Definition: Circular Permutations

The total number of unique ways to arrange a set of n distinct objects in a circle (*round table symmetry*) where absolute positions do not matter and only *relative order* counts is given by:

$$\text{Circular Arrangements} = \frac{n!}{n} = (n - 1)!$$

Intuitive Core: In a straight line, shifting everyone one seat to the right creates a completely new queue. Around a circular table, shifting everyone one seat to the right leaves their relative neighbors unchanged—the person to your left and right remains the same. To break this rotational symmetry, we must **anchor exactly one person** into a fixed seat. Once that reference point is locked down, the remaining $n - 1$ seats behave just like a standard linear row, falling into a $(n - 1)!$ countdown. **WARNING: DO NOT ALTER THE POOL TO $(N - 1)!$ IF THE SEATS ARE NUMBERED OR FIXED IN SPACE, AS THE ABSOLUTE POSITION RESTORES FULL LINEAR CODES $(N)!$**

Example: Arranging a Group Circularly

There are 6 distinct delegates attending an international conference. In how many unique relative ways can they be seated around a symmetrical round table?

Detailed Solution Steps:

1. Identify the total number of distinct individuals:

$$n = 6$$

2. Since the table is symmetrical and rotational shifts do not create unique relative configurations, choose 1 delegate to act as the spatial anchor.

3. Arrange the remaining $6 - 1 = 5$ delegates into the remaining open spots around the circle:

$$\text{Arrangements} = (6 - 1)! = 5!$$

4. Compute the final factorial value:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ ways}$$

Hint / Concept Review

Think of the anchor person as a landmark. They do not contribute to the counting choices because their job is simply to transform the circle into a straight line relative to themselves.

Once they take a seat, the other 5 seats can be filled as a standard row, resulting in $5! = 120$.

Subsection 1.15 Practice Exercises**Exercise 1.15.1**

A group of 5 friends are sitting down to have dinner at a round table. How many unique relative seating arrangements are possible?

- A. 5
- B. 24
- C. 120
- D. 720

Scratch Space:

Exercise 1.15.2

A committee consisting of 7 distinct members is meeting around a circular conference table.

- (a) Explain why a direct linear calculation of $7!$ overcounts the true number of unique relative configurations.
- (b) Calculate the absolute number of distinct circular permutations for this group.

Solution Space:

Exercise 1.15.3

Owen invites 3 guest speakers to join him at a round discussion table. If they take seats randomly around the table, how many unique relative configurations can be recorded by the meeting logs?

- A. 6
- B. 12
- C. 24
- D. 48

Scratch Space:

Exercise 1.15.4

A set of 6 unique keys is to be placed onto a simple circular metal key ring.

- (a) Given that flipping the key ring over makes clockwise and counter-clockwise views look identical, state how this extra layer of symmetry affects the $(n - 1)!$ base formula.
- (b) Calculate the total number of unique visual key arrangements on the ring.

Solution Space:

Exercise 1.15.5

A circular loop links $k + 1$ unique network servers together. Express the total number of unique relative tracking paths through these nodes in terms of k .

- A. $k!$
- B. $(k + 1)!$
- C. $k \times (k - 1)$
- D. $\frac{k!}{2}$

Scratch Space:

Exercise 1.15.6

An optimization tool evaluates seating patterns around a round table. The system tracks options for a group of n distinct managers.

- (a) Write out the algebraic factorial expression for the distinct relative seating layouts.
- (b) If the system logs exactly 720 unique relative configurations, determine the total number of managers n .

Solution Space:

Exercise 1.15.7

There are 4 boys and 4 girls sitting around a symmetrical circular table for a game[cite: 4]. If they can sit in any position, how many unique relative seating arrangements can be formed?

- A. 720
- B. 5040
- C. 40320
- D. 362880

Scratch Space:

Exercise 1.15.8

A school board meeting features 4 teachers from School A and 4 teachers from School B sitting around a circular table[cite: 2].

- (a) Find the total number of unique relative seating arrangements if the 4 teachers from School A must sit next to each other as a single block.
- (b) TRAP WARNING: A trainee suggests that because we group the 4 teachers from School A together, the remaining slots form a circle of 5 items, so the total choices are $5! \times 4!$. Explain the rotational error in this calculation.

Solution Space:

Exercise 1.15.9

A designer places 8 unique colored gemstones around the circular border of a custom luxury watch face. How many unique relative layouts can the manufacturing machine build?

- A. 720
- B. 5040
- C. 40320
- D. 362880

Scratch Space:

Exercise 1.15.10

An experimental tracking array connects m distinct calibration sensors into a closed ring loop.

- (a) State the basic circular permutation formula for the relative node layouts in terms of m .
- (b) Given that the layout registry logs exactly 24 unique relative configurations, solve for the sensor count parameter m .

Solution Space:

Exercise 1.15.11

A group of 6 people sit around a circular table. A student attempts to calculate the total unique ways to arrange them and answers $6! = 720$. This calculation is a failure because:

- A. Symmetrical table circles require using exponents.
- B. The linear formula $6!$ treats different rotations as separate layouts, ignoring the fact that shifting everyone around a circle doesn't alter their relative positions. We must divide by 6 or fix 1 anchor person, yielding $(6 - 1)! = 120$.
- C. The options must be processed using combinations.
- D. Relative paths cannot use factorial counting.

Scratch Space:

Exercise 1.15.12

An analyst evaluates security logs for token positions in a closed ring architecture. The system monitors a ring connecting 5 unique processing nodes.

- (a) Discuss the critical structural difference in counting pathways when nodes are arranged in a straight bus line versus a closed circular loop.
- (b) Perform the step-by-step calculation to show how anchoring one node scales the remaining arrangement options down to a standard factorial.

Solution Space:

2 Percentage

2.1 Basic Percentage Changes and Multipliers

“Percentages can be deceptive if we only look at what is being added or taken away. By shifting our perspective to see what remains, a percentage change transforms from an addition problem into a single multiplier step.”

Definition: Percentage Change and Growth Multipliers

A percentage change measures the relative difference between an initial value and a final value. Instead of calculating the change separately and adding or subtracting it, we convert the shift into a single growth or decay multiplier:

$$\text{New Value (Increase by } x\%) = \text{Original Value} \times (1 + x\%)$$

$$\text{New Value (Decrease by } x\%) = \text{Original Value} \times (1 - x\%)$$

Intuitive Core: Treating changes as multipliers keeps our calculations clean. An increase of 15% means the new amount is 115% of the baseline, which maps to a multiplier of 1.15. A decrease of 20% leaves 80% of the baseline behind, creating a multiplier of 0.80. **WARNING: ALWAYS APPLY THE PERCENTAGE MULTIPLIER DIRECTLY TO THE ORIGINAL BASE VALUE! NEVER MULTIPLY AN UNADJUSTED FINAL VALUE BY THE SAME SHIFT PERCENTAGE TO REVERSE THE CHANGE!**

Example: Applying Growth Multipliers Directly

The price of a mathematics text handbook is originally 180 HKD. If the price increases by 15% due to inflation, find the new retail price of the handbook.

Detailed Solution Steps:

1. Identify the original baseline value:

$$\text{Original Value} = 180 \text{ HKD}$$

2. Convert the 15% increase into a growth multiplier:

$$\text{Multiplier} = 1 + 15\% = 1 + 0.15 = 1.15$$

3. Calculate the adjusted price using a single multiplication step:

$$\text{New Price} = 180 \times 1.15 = 207 \text{ HKD}$$

Hint / Concept Review

Instead of calculating 15% of 180 (which is 27) and adding it back to 180, multiplying directly by 1.15 gets you straight to the final answer. This scale method keeps things simple when handling compound changes later on.

Subsection 2.1 Practice Exercises**Exercise 2.1.1**

The monthly enrollment of an arithmetic coaching center was 250 students last term. If the enrollment decreases by 12% this term, what is the new student enrollment?

- A. 30
- B. 218
- C. 220
- D. 280

Scratch Space:

Exercise 2.1.2

Owen owns a retail store. The import cost of a designer schoolbag is 450 HKD. He marks up the price of the bag by 40% for retail sale.

- (a) Write down the growth multiplier that represents a 40% markup.
- (b) Calculate the final retail price of the schoolbag.

Solution Space:

Exercise 2.1.3

A collection of test geometry sets has a baseline count of 80 sets. If the storage pool expands by 35%, find the updated total number of geometry sets available.

- A. 28
- B. 105
- C. 108
- D. 115

Scratch Space:

Exercise 2.1.4

The attendance at a school seminar drops from an initial count of 320 parents down to 240 parents in the second session.

- (a) Formulate the fraction equation required to determine the percentage change in attendance.
- (b) Calculate the exact percentage decrease.

Solution Space:

Exercise 2.1.5

An asset collection worth V dollars decreases in value by $x\%$. Express the updated worth of the asset pool using a single algebraic multiplier.

- A. $V - x$
- B. $V \times (1 - x)$
- C. $V \times \left(1 - \frac{x}{100}\right)$
- D. $\frac{V \cdot x}{100}$

Scratch Space:

Exercise 2.1.6

The length of a rectangular laboratory blueprint is L cm. The original length is increased by 20%, and its value becomes 72 cm.

- (a) Construct an algebraic equation using a single growth multiplier to find L .
- (b) Solve the equation to determine the original baseline length L .

Solution Space:

Exercise 2.1.7

A vintage luxury item is purchased for 5000 HKD. If its value increases by 8% every year, what is its value at the end of the first year?

- A. 5040 HKD
- B. 5400 HKD
- C. 5800 HKD
- D. 9000 HKD

Scratch Space:

}

Exercise 2.1.8

In an academic district tracking system, School A has 1200 registered students. The student population of School B is 15% less than that of School A .

- (a) Calculate the total student population of School B .
- (b) TRAP WARNING: A junior clerk claims that because School B has 15% fewer students than School A , School A must have exactly 15% more students than School B . Explain why this assumption is mathematically incorrect.

Solution Space:

Exercise 2.1.9

The volume of a gas decreases by 2.5% under pressure. If the original volume was 400 mL, what is the compressed volume?

- A. 375 mL
- B. 390 mL
- C. 397.5 mL
- D. 410 mL

Scratch Space:

Exercise 2.1.10

The price of a manufacturing component changes from a baseline value of k dollars to a final price of $k + 24$ dollars.

- (a) Write out an expression for the percentage increase in terms of k .
- (b) Given that the final price reflects an exact 12% increase over the baseline, calculate the value of k .

Solution Space:

Exercise 2.1.11

A smartphone price is reduced by 20% during a mid-year sale. A customer wants to find the original price from a sale price of 4000 HKD. They calculate $4000 \times 1.20 = 4800$ HKD. This calculation is a failure because:

- A. Percentage reductions cannot use decimals.
- B. The 20% discount applies to the unknown original price, not the sale price. To find the original price, the sale price must be divided by 0.80.
- C. The multiplier for a decrease should be added.
- D. Multipliers cannot reverse calculations.

Scratch Space:

Exercise 2.1.12

An analyst looks at a dataset where an initial index worth 150 points increases by 10% in the first tracking quarter, and then decreases by 10% in the second tracking quarter.

- (a) Discuss why the final value is not equal to the original 150 points, despite the identical upward and downward percentage shifts.
- (b) Compute the actual final index value to show the net effect of these consecutive changes.

Solution Space:

2.2 Profit, Loss, and Discount

“Financial math is built entirely on moving baselines. Profit and loss pivot around what an item cost to buy, while discounts pivot around the price printed on the tag. Keeping these baselines separate is the key to mastering commercial math.”

Definition: Commercial Financial Baselines

Every financial transaction operates across three distinct price points:

- **Cost Price (C)**: The original amount paid by the merchant to purchase or manufacture the item. This is the absolute baseline for profit and loss.
- **Marked Price (M)**: The initial advertised retail price printed on the price tag. This is the absolute baseline for discounts.
- **Selling Price (S)**: The final actual amount paid by the customer.

Intuitive Core: Profit or loss percentages always multiply the **Cost Price**. Discounts always multiply the **Marked Price**. Think of the Selling Price as a central meeting point that can be reached from two different directions:

$$\text{From Cost Price: } S = C \times (1 + \text{Profit}\%) \quad \text{or} \quad S = C \times (1 - \text{Loss}\%)$$

$$\text{From Marked Price: } S = M \times (1 - \text{Discount}\%)$$

WARNING: NEVER DEDUCT A DISCOUNT PERCENTAGE FROM THE COST PRICE, AND NEVER CALCULATE A PROFIT PERCENTAGE BASED ON THE MARKED PRICE!

Example: Bridging Cost Price and Marked Price

Owen buys a smartphone for a cost price of 3000 HKD. He marks up the item so that even after offering a 10% discount on the marked price, he still makes a 20% profit on the cost price. Find the marked price of the smartphone.

Detailed Solution Steps:

1. Calculate the Selling Price (S) from the Cost Price (C) baseline using the profit multiplier:

$$S = 3000 \times (1 + 20\%) = 3000 \times 1.20 = 3600 \text{ HKD}$$

2. Set up the equation reaching the same Selling Price (S) from the unknown Marked Price (M) using the discount multiplier:

$$S = M \times (1 - 10\%)$$

$$3600 = M \times 0.90$$

3. Solve for M using division:

$$M = \frac{3600}{0.90} = 4000 \text{ HKD}$$

Hint / Concept Review

Do not try to guess a single percentage change to jump straight from Cost Price to Marked Price.

Always use the Selling Price as a bridge. Calculate the exact dollar value of the Selling Price first, then reverse the discount to find the marked price on the tag.

Subsection 2.2 Practice Exercises**Exercise 2.2.1**

A merchant buys an item for 400 HKD and sells it for 520 HKD. What is the profit percentage made on the transaction?

- A. 23%
- B. 25%
- C. 30%
- D. 130%

Scratch Space:

Exercise 2.2.2

The marked price of a scientific calculator is 250 HKD. Owen's store offers a 16% discount during a back-to-school promotional campaign.

- (a) Formulate the equation using a discount multiplier to determine the actual selling price.
- (b) Calculate the final selling price of the calculator.

Solution Space:

Exercise 2.2.3

An item is sold for 720 HKD, resulting in a 10% loss for the shop. Find the original cost price paid by the merchant to stock the item.

- A. 648 HKD
- B. 792 HKD
- C. 800 HKD
- D. 820 HKD

Scratch Space:

Exercise 2.2.4

The cost price of a digital watch is 600 HKD. The watch is marked up by 50% to set its initial marked price, and it is later sold at a 20% discount on its marked price.

- (a) Find the marked price and the final selling price of the digital watch.
- (b) Calculate the final profit percentage made based on the initial cost price.

Solution Space:

Exercise 2.2.5

A business sells a product for S dollars, making a profit of $g\%$. Express the cost price of the product using a simplified algebraic multiplier in terms of S and g .

A. $S \times (1 - g\%)$

B. $\frac{S}{1+g\%}$

C. $S \times (1 + g\%)$

D. $\frac{S \cdot g}{100}$

Scratch Space:

Exercise 2.2.6

The marked price of a leather jacket is M dollars. A store clerk applies a discount of $d\%$ to set the final selling price.

- (a) Construct an algebraic expression for the selling price using a single decay multiplier.
- (b) If the jacket's marked price is 1500 dollars and it sells for 1200 dollars, determine the discount percentage d .

Solution Space:

Exercise 2.2.7

A bookshop sells a novel at a discount of 20%, making a profit of 12% on the cost price[cite: 4]. If the cost price of the novel is 80 HKD, find its marked price.

- A. 89.6 HKD
- B. 100 HKD
- C. 112 HKD
- D. 114 HKD

Scratch Space:

Exercise 2.2.8

A boutique marks up its clothing line by 60% over the cost price to set the retail tag price[cite: 4]. During a seasonal clearance, the store advertises a 30% discount on the marked price[cite: 4].

- (a) Find the selling price of an item expressed as a percentage of its cost price.
- (b) TRAP WARNING: A sales assistant mistakenly tells a customer that a 60% markup followed by a 30% discount results in a net profit of exactly $60\% - 30\% = 30\%$. Mathematically expose the flaw in this statement by calculating the actual net profit percentage.

Solution Space:

Exercise 2.2.9

If an item is sold for 960 HKD after a 20% discount is applied to its marked price, what would the selling price be if it were sold at a 5% discount instead?

- A. 1140 HKD
- B. 1152 HKD
- C. 1200 HKD
- D. 1260 HKD

Scratch Space:

Exercise 2.2.10

A manufacturer produces a machinery part for a cost price of C dollars. The marked price is structured such that if a 15% discount is granted, the company still locks in a profit of 36% over production costs.

- (a) Write an equation linking C and the marked price M using their corresponding multipliers.
- (b) Given that the marked price is set to 640 dollars, evaluate the original cost price C .

Solution Space:

Exercise 2.2.11

A bicycle is sold for 2400 HKD at a discount of 20% off the marked price. A student wants to find the marked price and calculates $2400 \times 1.20 = 2880HKD$. This calculation is a failure because:

- A. Financial math cannot use decimal approximations.
- B. The 20% discount is a reduction based on the unknown marked price, not an increase on the selling price. To reverse the discount, the selling price must be divided by 0.80, yielding 3000 HKD.
- C. The profit baseline must override the discount tag.
- D. Multipliers cannot divide monetary values.

Scratch Space:

Exercise 2.2.12

An auditor reviews a merchant's inventory ledger. The ledger records a batch of items that were sold at a nominal discount of 25% off the marked price, yet the business still managed to record a 5% net profit on their initial cost price.

- (a) Discuss how a business can offer a large discount on a price tag while maintaining a positive net profit margin over structural costs.
- (b) Calculate the explicit markup percentage that was originally applied to the cost price to generate the marked price tag.

Solution Space:

2.3 Successive Percentage Changes

“When an amount changes multiple times in a row, the baseline shifts with every step. We cannot simply add or subtract the percentages; we must chain the multipliers together so that each change builds on top of the last.”

Theorem: Successive Percentage Changes

If an initial quantity V_0 undergoes a series of consecutive percentage changes ($x_1\%$, $x_2\%$, $\dots$, $x_n\%$), each subsequent change applies directly to the newly accumulated value. The final value V_n is computed by multiplying the original value by the entire chain of individual growth or decay multipliers:

$$V_n = V_0 \times (1 \pm x_1\%) \times (1 \pm x_2\%) \times \cdots \times (1 \pm x_n\%)$$

Intuitive Core: Successive changes create a compounding effect. If a stock value increases by 20% and then decreases by 20%, the net result is not a 0% change. The first shift scales the value to 1.20, and the second shift scales that new result by 0.80. Chaining them gives a total multiplier of $1.20 \times 0.80 = 0.96$, showing a net loss of 4%. **WARNING: NEVER ADD OR SUBTRACT SUCCESSIVE PERCENTAGE RATES DIRECTLY! ALMOST EVERY EXAM TRAP IS DESIGNED TO EXPLOIT THIS INCORRECT SHORTCUT!**

Example: Chaining Consecutive Shifts

An investment fund worth 10000 HKD grows by 10% in the first year, but drops by 20% in the second year due to a market correction. Find the final value of the investment fund at the end of the second year.

Detailed Solution Steps:

1. Identify the original baseline asset value:

$$V_0 = 10000 \text{ HKD}$$

2. Establish the separate multipliers for each stage:

$$\text{Multiplier 1 (Increase by 10\%)} = 1 + 0.10 = 1.10$$

$$\text{Multiplier 2 (Decrease by 20\%)} = 1 - 0.20 = 0.80$$

3. Chain the multipliers together to calculate the final value:

$$V_2 = 10000 \times 1.10 \times 0.80$$

$$V_2 = 11000 \times 0.80 = 8800 \text{ HKD}$$

Hint / Concept Review

Notice that a common trap is to assume the net change is $+10\% - 20\% = -10\%$, which would give an incorrect answer of 9000 HKD.

By chaining the multipliers ($1.10 \times 0.80 = 0.88$), you instantly see that the fund is left with exactly 88% of its original baseline value, yielding 8800 HKD.

Subsection 2.3 Practice Exercises**Exercise 2.3.1**

The population of a small district increases by 5% in the first year and then increases by 10% in the second year. If the initial population was 20000, what is the population at the end of the second year?

- A. 23000
- B. 23100
- C. 23200
- D. 24000

Scratch Space:

Exercise 2.3.2

The price of a certain corporate stock is 50 HKD. Owen monitors the market and notes that the price increases by 20% on Monday but drops by 15% on Tuesday.

- (a) Write out the chained multiplication expression used to track the stock price across both days.
- (b) Calculate the final price of the stock at the end of Tuesday.

Solution Space:

Exercise 2.3.3

A factory's manufacturing output decreases by 10% in the first quarter and further decreases by 30% in the second quarter. Find the overall percentage decrease in manufacturing output over these two quarters combined.

- A. 37%
- B. 40%
- C. 63%
- D. 70%

Scratch Space:

Exercise 2.3.4

The entry fee for an exhibition is marked up by 25% over its baseline rate. Due to a decline in visitors, the management introduces a 20% discount on the marked-up price.

- (a) Use a chain of multipliers to determine the final entry fee expressed as a percentage of the baseline rate.
- (b) Deduce the net overall percentage change in the entry fee from start to finish.

Solution Space:

Exercise 2.3.5

An initial value A is increased by $x\%$ and then the resulting value is decreased by $y\%$. Express the final value using a simplified algebraic product array in terms of A , x , and y .

A. $A \times \left(1 + \frac{x-y}{100}\right)$

B. $A \times \left(1 + \frac{x}{100}\right) \times \left(1 - \frac{y}{100}\right)$

C. $A \times \left(\frac{x \cdot y}{10000}\right)$

D. $A + \frac{x}{100} - \frac{y}{100}$

Scratch Space:

Exercise 2.3.6

The value of a server network index is I . The index increases by 15% in the first month, and then increases by $k\%$ in the second month. The combined effect of these two changes is a net total increase of 38% over the original index I .

- (a) Construct an algebraic equation matching this compound growth pattern using chained multipliers.
- (b) Solve the equation to determine the exact percentage value of k .

Solution Space:

Exercise 2.3.7

A real estate property increases in value by 8% in 2024 and increases by another 12% in 2025. What is the overall compound percentage increase in the property value over these two years combined?

- A. 19.6%
- B. 20%
- C. 20.96%
- D. 21.6%

Scratch Space:

Exercise 2.3.8

An electrical retail store marks up a laptop computer by 40% over its original production cost price. To attract customers during a holiday sale, the manager applies a succession of two identical discounts of 20% one after the other on the marked price.

- (a) Find the final selling price of the laptop expressed as a percentage of its original cost price.
- (b) TRAP WARNING: A sales clerk tells a buyer that a 40% markup followed by two 20% discounts results in a net loss of exactly $40\% - 20\% - 20\% = 0\%$, meaning the shop breaks even. Show mathematically why this explanation is wrong, and state whether the shop makes a profit or a loss.

Solution Space:

Exercise 2.3.9

An item is subjected to a 10% increase followed by a 10% decrease. If the final selling price is 990 HKD, what was the original price before these changes took place?

- A. 990 HKD
- B. 1000 HKD
- C. 1010 HKD
- D. 1100 HKD

Scratch Space:

Exercise 2.3.10

The hourly wage of a factory supervisor is W dollars. The wage increases by 8% due to a promotion, but later decreases by 5% when the factory downscales operational hours.

- (a) Write a simplified expression for the final hourly wage in terms of W .
- (b) Given that the final hourly wage tracks out to exactly 102.60 dollars, calculate the initial baseline wage value W .

Solution Space:

Exercise 2.3.11

The price of a digital camera is increased by 30% and then later decreased by 30%. A student states that because the percentage rates are identical, the final price remains unchanged. This reasoning is a failure because:

- A. Percentage values cannot be applied twice to monetary assets.
- B. The 30% decrease is calculated based on the new, larger marked-up price, which removes more value than the initial increase added.
- C. Multipliers must be added instead of chained.
- D. Reversing successive changes requires using a negative base.

Scratch Space:

Exercise 2.3.12

An economic analyst tracks a retail market index over two consecutive quarters. In the first quarter, the index drops by 20%. In the second quarter, it recovers by increasing by 25% based on its end-of-first-quarter level.

- (a) Discuss how a lower baseline value at the start of the second quarter allows a larger percentage increase to exactly balance out a smaller initial percentage drop.
- (b) Perform the full multiplier chain step to prove that the index value at the end of the second quarter is completely identical to its original starting value.

Solution Space:

2.4 Compound Interest and Growth

“Interest that compounds is a runaway train powered by time. Because the interest earned at the end of each period is folded back into the principal, the baseline grows continuously, turning simple growth into an exponential curve.”

Theorem: The Compound Interest Formula

When a principal amount P is invested at an interest rate of $r\%$ per period for n consecutive compounding periods, the interest earned at the end of each stage is accumulated directly into the balance. The final accumulated amount A and the total compound interest earned I are given by:

$$A = P \times (1 + r\%)^n$$

$$I = A - P$$

Intuitive Core: Compound interest is simply a specific case of successive percentage changes where the growth rate remains completely constant over time. Multiplying by the same growth multiplier $(1+r\%)$ over n consecutive periods condenses the operation into an exponential power structure. **WARNING: ALWAYS MATCH THE INTEREST RATE TO THE EXACT DURATION OF THE COMPOUNDING PERIOD! IF AN ANNUAL INTEREST RATE COMPOUNDS MONTHLY, YOU MUST DIVIDE THE RATE BY 12 AND MULTIPLY THE TOTAL YEARS BY 12 BEFORE COMPUTING THE EXPONENT!**

Example: Compounding Interest Across Sub-Periods

Owen deposits 20000 HKD into a savings account that offers an annual interest rate of 6% compounded semi-annually. Find the total accumulated amount in his account at the end of 2 years.

Detailed Solution Steps:

1. Identify the principal investment:

$$P = 20000 \text{ HKD}$$

2. Adjust the annual interest rate to match the semi-annual period:

$$\text{Period Rate } r\% = \frac{6\%}{2} = 3\% = 0.03$$

3. Calculate the total number of compounding periods n across 2 years:

$$n = 2 \text{ years} \times 2 \text{ periods/year} = 4 \text{ periods}$$

4. Apply the exponential compound growth formula:

$$A = 20000 \times (1 + 3\%)^4$$

$$A = 20000 \times (1.03)^4 \approx 22510.18 \text{ HKD}$$

Hint / Concept Review

Because interest compounds semi-annually (twice a year), the bank calculates interest every 6 months.

Instead of applying a 6% multiplier 2 times, you must apply a smaller 3% multiplier 4 times over the 2-year timeline.

Subsection 2.4 Practice Exercises**Exercise 2.4.1**

A sum of 8000 HKD is invested at an annual interest rate of 4% compounded annually. What is the total accumulated amount at the end of 3 years?

- A. 8960.00 HKD
- B. 8998.91 HKD
- C. 9000.30 HKD
- D. 9358.88 HKD

Scratch Space:

Exercise 2.4.2

Owen deposits a principal of 50000 HKD into a high-yield bank account that bears an annual interest rate of 8% compounded quarterly.

- (a) Determine the adjusted interest rate per compounding period and the total number of periods over 1 year.
- (b) Calculate the total compound interest earned by Owen at the end of 1 year.

Solution Space:

Exercise 2.4.3

The population of a city is growing exponentially at a constant rate of 5% per year. If the current population is 600000, what will the estimated population be after 4 years?

- A. 720000
- B. 729304
- C. 730500
- D. 765769

Scratch Space:

Exercise 2.4.4

A business invests 15000 HKD into a commercial venture. The value of the investment compounds annually at a fixed rate. At the end of 2 years, the total accumulated value of the asset reaches 17496 HKD.

- (a) Construct an exponential equation that models this compound growth track.
- (b) Solve the equation to determine the annual growth rate percentage.

Solution Space:

Exercise 2.4.5

A principal amount P is deposited at an annual interest rate of $r\%$ compounded monthly for t years. Express the total accumulated amount using a simplified exponential algebraic expression.

- A. $P \times (1 + r\%)^{12t}$
- B. $P \times \left(1 + \frac{r}{1200}\right)^{12t}$
- C. $P \times (1 + 12r\%)^t$
- D. $P + P \cdot \left(\frac{r \cdot t}{1200}\right)$

Scratch Space:

Exercise 2.4.6

The population of a tracked bacterial culture grows exponentially by $x\%$ every hour. Initially, the culture contains 4000 active bacteria. After 3 hours, the population expands to exactly 6912 bacteria.

- (a) Construct an algebraic expression that isolates the compound multiplier using the given parameters.
- (b) Solve for the hourly percentage growth rate x .

Solution Space:

Exercise 2.4.7

A student invests 10000 HKD at an annual interest rate of 6% compounded semi-annually[cite: 4]. What is the total compound interest earned at the end of 2 years?

- A. 1200.00 HKD
- B. 1255.09 HKD
- C. 1261.62 HKD
- D. 11255.09 HKD

Scratch Space:

Exercise 2.4.8

A multi-national asset group deposits 100000 HKD into a corporate bond scheme that yields a fixed nominal interest rate of 12% per annum[cite: 4]. The contract allows the client to choose between annual compounding or monthly compounding.

- (a) Calculate the total accumulated values at the end of 1 year under both compounding options.
- (b) TRAP WARNING: A junior account representative claims that because the nominal annual percentage rate remains exactly 12% in both packages, the final interest payout will be identical. Mathematically disprove this claim by evaluating the exact dollar difference between the two choices.

Solution Space:

Exercise 2.4.9

An engineering fund accumulates value according to the compound growth model. If a deposit of 5000 HKD grows to 6050 HKD after 2 years of annual compounding, what will the total value of the deposit be at the end of 3 years?

- A. 6555.00 HKD
- B. 6600.00 HKD
- C. 6655.00 HKD
- D. 7320.50 HKD

Scratch Space:

Exercise 2.4.10

A tracking portfolio models an investment value that expands according to the compound interest formula. A principal of P dollars is invested at an annual interest rate of $r\%$ compounded annually. After 20 years, the total accumulated amount is found to be exactly double the initial principal ($2P$).

- (a) Set up the exponential base equation matching this doubling condition.
- (b) Formulate the specific expression needed to solve for the annual interest rate r using radical roots.

Solution Space:

Exercise 2.4.11

A student wants to calculate the final value of a 3000 HKD investment after 2 years at an annual interest rate of 10% compounded semi-annually. They apply the calculation $3000 \times (1 + 0.10)^4 = 4392.30$ HKD. This calculation is a failure because:

- A. Interest calculations cannot use integer exponents.
- B. The student multiplied the number of compounding periods but forgot to divide the annual interest rate by 2, causing the equation to apply a massive 10% rate per period instead of the correct 5% rate.
- C. Semi-annual growth requires using a linear product.
- D. The principal must be subtracted from the multiplier base.

Scratch Space:

Exercise 2.4.12

An economic advisor analyzes a long-term development fund tracking system. A sum of 40000 HKD is split into two independent profiles. Profile X grows at 6% simple interest per annum, while Profile Y grows at 6% compound interest compounded annually.

- (a) Compare the total value profiles of both investments at the end of the first year and explain why they match.
- (b) Calculate and contrast their accumulated values at the end of 3 years to illustrate how compound interest outperforms simple linear growth over time.

Solution Space:

2.5 Depreciation and Decay

“As machinery ages or technology advances, asset values slip away. Exponential decay behaves like compound interest in reverse—with every passing period, the loss applies to a shrinking base, slowing down the asset’s drop in value over time.”

Theorem: The Depreciation Formula

When an initial asset value P depreciates at a constant rate of $r\%$ per period for n consecutive compounding periods, the updated book value A and the total value lost D are given by:

$$A = P \times (1 - r\%)^n$$

$$D = P - A$$

Intuitive Core: Depreciation is simply successive percentage decreases where the rate of decay stays identical across each time interval. Multiplying by a constant decay multiplier $(1 - r\%)$ over n periods creates an **exponential decay structure**. Because the base decreases at each step, the absolute amount of value lost also decreases each period. **WARNING: NEVER CALCULATE TOTAL DEPRECIATION BY MULTIPLYING THE RATE BY THE NUMBER OF YEARS DIRECTLY ($P \times r\% \times n$), AS THIS INCORRECTLY TREATS EXPOENTIAL DECAY AS SIMPLE LINEAR LOSS!**

Example: Calculating Exponential Asset Decay

Owen buys a commercial laser printer for his retail store at an initial cost of 16000 HKD. If the printer’s value depreciates at a constant rate of 10% per annum, find its depreciated book value at the end of 3 years.

Detailed Solution Steps:

1. Identify the initial baseline cost:

$$P = 16000 \text{ HKD}$$

2. Convert the 10% depreciation rate into a constant decay multiplier:

$$\text{Multiplier} = 1 - 10\% = 1 - 0.10 = 0.90$$

3. Apply the exponential depreciation formula for $n = 3$ years:

$$A = 16000 \times (0.90)^3$$

$$A = 16000 \times 0.729 = 11664 \text{ HKD}$$

Hint / Concept Review

At the end of each year, the printer retains exactly 90% of the value it had at the beginning of that year.

Instead of calculating a 10% reduction year-by-year manually, raising the multiplier to the power of 3 ($0.90^3 = 0.729$) gives the final remaining value factor instantly.

Subsection 2.5 Practice Exercises**Exercise 2.5.1**

A company vehicle is purchased for 250000 HKD. If it depreciates at a constant rate of 20% per year, what is its book value at the end of 2 years?

- A. 150000 HKD
- B. 160000 HKD
- C. 190000 HKD
- D. 200000 HKD

Scratch Space:

Exercise 2.5.2

Owen updates his office computers. The initial value of the server hardware network is 80000 HKD. It is given that the hardware value depreciates at a constant rate of 15% per annum.

- (a) Formulate the exponential expression used to determine the remaining value of the servers after 3 years.
- (b) Calculate the total value lost by the server network over this 3-year period.

Solution Space:

Exercise 2.5.3

The population of a rural village is decreasing exponentially at a constant rate of 4% per year due to urban migration. If the initial population was 12500, what will the population be after 2 years?

- A. 11500
- B. 11520
- C. 11540
- D. 12000

Scratch Space:

Exercise 2.5.4

A piece of industrial factory equipment is bought for 50000 HKD. Its value depreciates annually at a fixed rate. At the end of 2 years, its book value drops to exactly 40500 HKD.

- (a) Construct an exponential equation that models this annual loss of value.
- (b) Solve the equation to find the annual depreciation rate percentage.

Solution Space:

Exercise 2.5.5

An asset with an initial value of V dollars depreciates at an annual rate of $r\%$ compounded annually. Express the total value lost by the asset after t years using a simplified algebraic expression.

A. $V \times \left(1 - \frac{r}{100}\right)^t$

B. $V - V \times \left(1 - \frac{r}{100}\right)^t$

C. $V \cdot \left(\frac{r \cdot t}{100}\right)$

D. $V \times \left(\frac{r}{100}\right)^t$

Scratch Space:

Exercise 2.5.6

The concentration of a volatile chemical compound in a testing container evaporates exponentially by $x\%$ every hour. Initially, the solution holds 2000 mg of the compound. After 3 hours, the remaining mass drops to exactly 1024 mg.

- (a) Construct an algebraic equation to isolate the constant decay multiplier using the given values.
- (b) Solve for the hourly percentage decay rate x .

Solution Space:

Exercise 2.5.7

A luxury wristwatch was purchased for 60000 HKD. If its value depreciates at a constant rate of 5% per year[cite: 4], what is the total value lost by the wristwatch at the end of 2 years?

- A. 5700.00 HKD
- B. 5850.00 HKD
- C. 6000.00 HKD
- D. 54150.00 HKD

Scratch Space:

Exercise 2.5.8

An engineering plant tracks the value of two machines purchased at the same time[cite: 4]. Machine A is purchased for 100000 HKD and loses value under a fixed linear model at a rate of 10000 HKD flat per year. Machine B is purchased for 100000 HKD and loses value under a constant depreciation model at a rate of 10% compounded annually.

- (a) Compute the book value of both machines at the end of 2 years.
- (b) TRAP WARNING: A production tech claims that because both machines lose exactly 10% of their initial purchase cost in the first year, they will have identical book values at the end of any year. Mathematically disprove this claim by evaluating the book values at the end of year 3.

Solution Space:

Exercise 2.5.9

The value of an electronic component drops according to exponential decay. If a piece of hardware costing 4000 HKD depreciates to 3240 HKD after 2 years of annual compounding, what will its book value be at the end of 3 years?

- A. 2800.00 HKD
- B. 2916.00 HKD
- C. 2940.00 HKD
- D. 3000.00 HKD

Scratch Space:

Exercise 2.5.10

A tracking portfolio models the value of factory machinery that shrinks according to the depreciation formula. An initial value of P dollars depreciates at a constant annual rate of $r\%$ compounded annually. After 15 years, the remaining book value is found to be exactly half of the initial cost ($\frac{1}{2}P$).

- (a) Set up the exponential base equation matching this halving condition.
- (b) Formulate the specific expression needed to solve for the annual depreciation rate r using radical roots.

Solution Space:

Exercise 2.5.11

An office projector costing 5000 HKD depreciates at a constant annual rate of 20% compounded annually. A student wants to find the value lost after 2 years and calculates $5000 \times (0.20)^2 = 200$ HKD. This calculation is a failure because:

- A. Depreciation values cannot use exponents.
- B. The student raised the depreciation rate itself to a power instead of using the decay multiplier $(1 - 0.20) = 0.80$, which calculates the remaining value rather than the raw loss.
- C. The baseline value should be added.
- D. Multipliers cannot process asset loss profiles.

Scratch Space:

Exercise 2.5.12

An industrial auditor reviews the depreciation logs of an asset tracking system over a 3-year timeline. A machine purchased for 120000 HKD loses value at an annual rate of 15% compounded annually.

- (a) Discuss why the absolute amount of value lost in the third year is smaller than the absolute value lost in the first year.
- (b) Complete the step-by-step mathematical breakdown for all 3 years to verify the exact book value remaining at the end of the timeline.

Solution Space:

3 Probability

3.1 Classical Definition of Probability

“Probability is the mathematical tool we use to measure uncertainty. By comparing the specific outcomes we are looking for against the entire pool of what could possibly happen, we turn a blind guess into a precise ratio.”

Definition: Classical Probability

If an experiment has a finite *sample space* S containing N unique, equally likely outcomes, and an event A corresponds to a specific sub-collection of $n(A)$ outcomes from that pool, then the probability of event A occurring, denoted by $P(A)$, is defined as the ratio:

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

By definition, the value of any probability is bounded such that $0 \leq P(A) \leq 1$.

Intuitive Core: Probability is a fraction that measures structural strength. A probability of 0 means the event is an absolute impossibility, while a probability of 1 means it is an absolute certainty. To calculate any basic probability, you must perform two completely independent counting tasks: find your denominator first by counting the total possible states, then find your numerator by counting the specific states that satisfy the question. **WARNING: THE CLASSICAL FORMULA ONLY APPLIES IF EVERY SINGLE INDIVIDUAL OUTCOME IN THE POOL HAS THE EXACT SAME BASELINE STRENGTH OR CHANCE OF OCCURRING!**

Example: Evaluating Ratios from Finite Pools

A box contains 7 red balls, 5 blue balls, and 8 white balls[cite: 4]. If one ball is randomly drawn from the box, find the probability that the chosen ball is blue.

Detailed Solution Steps:

1. Count the total number of possible outcomes in the sample space $n(S)$:

$$n(S) = 7 + 5 + 8 = 20 \text{ outcomes}$$

2. Count the number of favorable outcomes matching the event $n(\text{Blue})$:

$$n(\text{Blue}) = 5 \text{ outcomes}$$

3. Apply the classical probability ratio formula:

$$P(\text{Blue}) = \frac{n(\text{Blue})}{n(S)} = \frac{5}{20}$$

4. Simplify the resulting fraction:

$$P(\text{Blue}) = \frac{1}{4} = 0.25$$

Hint / Concept Review

Think of this as 5 target slots out of 20 total possibilities.

Because only a single item is being pulled, no complex permutations or arrangements are needed. Your target pool sits directly on top of your universe pool.

Subsection 3.1 Practice Exercises**Exercise 3.1.1**

A card is drawn at random from a standard deck of 52 playing cards. What is the probability that the drawn card is a King?

- A. $\frac{1}{52}$
- B. $\frac{1}{13}$
- C. $\frac{1}{4}$
- D. $\frac{4}{13}$

Scratch Space:

Exercise 3.1.2

A standard fair six-sided die is rolled once.

- (a) List all the favorable prime number outcomes that can appear on the upper face.
- (b) Calculate the probability of rolling a prime number.

Solution Space:

Exercise 3.1.3

A box contains ten cards numbered 2, 3, 5, 8, 11, 11, 12, 15, 19, and 24 respectively[cite: 4]. If a card is randomly drawn from the box, find the probability that the number on the card is a multiple of 3[cite: 4].

- A. $\frac{1}{10}$
- B. $\frac{3}{10}$
- C. $\frac{2}{5}$
- D. $\frac{1}{2}$

Scratch Space:

Exercise 3.1.4

Owen has a structural bag containing 8 white socks, 4 yellow socks, and 2 red socks[cite: 4]. He shakes the bag and pulls out exactly one sock at random[cite: 4].

- (a) Determine the probability that Owen draws a red sock.
- (b) Find the probability that the chosen sock is either white or yellow.

Solution Space:

Exercise 3.1.5

A raffle drum contains x premium tickets and $3x$ basic tickets. If an inspector extracts exactly one ticket at random, express the probability of drawing a premium ticket in terms of a simplified algebraic fraction.

- A. $\frac{1}{x}$
- B. $\frac{1}{3}$
- C. $\frac{1}{4}$
- D. $\frac{3}{4}$

Scratch Space:

Exercise 3.1.6

A storage locker holds a collection of n standard math calculators and $2n-2$ financial calculators. A teacher reaches in and picks one calculator profile at random.

- (a) Construct an algebraic expression representing the probability of picking a standard math calculator.
- (b) Given that the probability of choosing a standard math calculator is exactly $\frac{2}{5}$, determine the total number of calculators in the locker.

Solution Space:

Exercise 3.1.7

There are 18 boys and 12 girls in a class. If a student is randomly selected from the class to act as a committee representative, find the probability that the selected student is a girl.

- A. $\frac{2}{3}$
- B. $\frac{3}{5}$
- C. $\frac{3}{4}$
- D. $\frac{1}{30}$

Scratch Space:

Exercise 3.1.8

An examination database stores the performance records of 200 candidates[cite: 4]. The files indicate that 140 candidates are male and 60 candidates are female[cite: 3]. If the network automated picker selects exactly one file entry at random,

- (a) Evaluate the probability that the selected record belongs to a male candidate.
- (b) TRAP WARNING: A data clerk argues that because the file is split into two distinct gender blocks, the baseline probability of selecting a female file must be exactly $\frac{1}{2}$. Explain the mathematical error in this reasoning.

Solution Space:

Exercise 3.1.9

A two-digit integer is randomly selected from the set of all integers from 10 to 99 inclusive. Find the probability that the chosen number is a perfect square.

- A. $\frac{1}{15}$
- B. $\frac{7}{90}$
- C. $\frac{2}{25}$
- D. $\frac{1}{10}$

Scratch Space:

Exercise 3.1.10

A custom lottery box contains a set of tracking tokens. There are k^2 blue tokens and $7k$ white tokens. A token is pulled out at random.

- (a) State the probability tracking formula for drawing a white token in terms of k .
- (b) Given that the probability of drawing a white token scales to exactly $\frac{7}{11}$, calculate the structural metric value k .

Solution Space:

Exercise 3.1.11

A bag contains 3 red marbles and 2 blue marbles. A student wants to find the probability of drawing a red marble and writes down the answer as $\frac{3}{2} = 1.5$. This evaluation is a failure because:

- A. Classical ratios can only use odd numbers.
- B. A probability value can never exceed 1; the numerator must be divided by the total sample space pool ($3 + 2 = 5$), yielding $\frac{3}{5}$.
- C. Marbles require a permutation counter.
- D. The numbers must be multiplied by their sample size.

Scratch Space:

Exercise 3.1.12

An analyst evaluates an automated factory screening system. A production bin holds 45 fully functional components and 5 defective components. The robotic arm picks exactly one component profile at random.

- (a) Discuss the foundational conditions that must be guaranteed in the physical design to ensure that classical probability ratios accurately predict choices.
- (b) Map out the fractional calculation to prove the exact probability of picking a non-defective component.

Solution Space:

3.2 Addition Law of Probability and Mutually Exclusive Events

“When we look for outcomes that satisfy one event or another, we are combining separate targets. If these targets cannot happen at the same time, finding their combined strength is as simple as adding their individual paths.”

Definition: Mutually Exclusive Events

Two events A and B are said to be mutually exclusive if they cannot occur simultaneously. In the language of sets, their intersection is empty ($A \cap B = \emptyset$), which means the probability of both occurring at the same time is exactly zero:

$$P(A \text{ and } B) = 0$$

Theorem: The Addition Law for Mutually Exclusive Events

If two events A and B are mutually exclusive, then the probability that event A *or* event B occurs is equal to the sum of their individual probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

Intuitive Core: We apply the Addition Law when our compound condition is joined by the word “OR”. If the events are mutually exclusive, their individual paths are completely separate pools with no overlap. Therefore, you can add their probabilities directly. **WARNING: NEVER ADD PROBABILITIES DIRECTLY IF IT IS POSSIBLE FOR BOTH EVENTS TO OCCUR AT THE SAME TIME! IF AN OVERLAP EXISTS, YOU MUST SWITCH TO THE GENERAL ADDITION LAW AND SUBTRACT THE INTERSECTION!**

Example: Combining Non-Overlapping Probabilities

A fair six-sided die is rolled once. Find the probability that the score on the upper face is either a 1 *or* a prime number.

Detailed Solution Steps:

1. Define the individual event pools:

$$\text{Event } A = \{1\} \implies P(A) = \frac{1}{6}$$

$$\text{Event } B = \{2, 3, 5\} \implies P(B) = \frac{3}{6}$$

2. Check if the events are mutually exclusive: Can a face score be both 1 and a prime number simultaneously? No, since 1 is not a prime number.

3. Apply the Addition Law directly:

$$P(A \text{ or } B) = P(A) + P(B) = \frac{1}{6} + \frac{3}{6}$$

$$P(A \text{ or } B) = \frac{4}{6} = \frac{2}{3} \approx 0.667$$

Hint / Concept Review

Identify the keyword *or*. Since the face cannot be 1 and prime at the same time, the categories are mutually exclusive.

Adding their fractional probabilities gives the correct total target strength directly.

Subsection 3.2 Practice Exercises**Exercise 3.2.1**

A card is drawn at random from a standard deck of 52 playing cards. What is the probability that the drawn card is either a King *or* a Queen?

- A. $\frac{1}{52}$
- B. $\frac{1}{26}$
- C. $\frac{2}{13}$
- D. $\frac{4}{13}$

Scratch Space:

Exercise 3.2.2

A bag contains 5 red balls, 6 yellow balls, and 3 white balls[cite: 4]. One ball is randomly drawn from the bag.

- (a) State why the events “drawing a red ball” and “drawing a white ball” are mutually exclusive.
- (b) Calculate the probability that the drawn ball is either red *or* white.

Solution Space:

Exercise 3.2.3

The probability that a student passes a math quiz is 0.75, and the probability that the student scores a perfect mark is 0.12. If these two outcomes are nested parts of the same performance grading scale, find the probability that a student fails the quiz.

- A. 0.13
- B. 0.25
- C. 0.63
- D. 0.87

Scratch Space:

Exercise 3.2.4

Owen operates a local boutique store. The probability that a customer pays using credit card is 0.45, the probability that they pay using an e-wallet app is 0.35, and the remaining customers pay using cash. Assume each customer uses exactly one payment channel.

- (a) Determine the probability that a randomly selected customer pays using either a credit card *or* an e-wallet app.
- (b) Calculate the probability that a customer pays using cash.

Solution Space:

Exercise 3.2.5

A collection of items is split into three distinct, non-overlapping subsets: A , B , and C . The probability of picking an item from A is $2x$, from B is $3x$, and from C is $4x$. If there are no other items in the collection, express the value of x .

- A. $\frac{1}{9}$
- B. $\frac{1}{5}$
- C. $\frac{1}{3}$
- D. 1

Scratch Space:

Exercise 3.2.6

An automated script routes incoming data packets into three mutually exclusive channels: Channel X , Channel Y , and Channel Z . The probability that a packet is routed to Channel X is $k - 0.1$, and the probability it goes to Channel Y is $2k$.

- (a) Write an algebraic expression for the probability that a packet is routed to either Channel X or Channel Y .
- (b) Given that the probability of routing to Channel Z is 0.4, and every packet must be routed to one of the three channels, solve for the parameter k .

Solution Space:

Exercise 3.2.7

A box contains cards numbered 1 to 20. If one card is randomly drawn, find the probability that the number on the card is either a multiple of 7 *or* a multiple of 9.

- A. $\frac{1}{10}$
- B. $\frac{3}{20}$
- C. $\frac{1}{5}$
- D. $\frac{1}{4}$

Scratch Space:

Exercise 3.2.8

An exam coordinator compiles the distribution charts for a cohort of students[cite: 4]. The probability that a student achieves a top grade of Level 5** is 0.05, and the probability of achieving Level 5* is 0.11[cite: 4].

- (a) Evaluate the probability that a randomly chosen student achieves either Level 5** *or* Level 5*.
- (b) TRAP WARNING: A teacher argues that if we want to find the probability of a student failing to achieve a Level 5** or Level 5*, we can calculate it by evaluating $(1 - 0.05) + (1 - 0.11)$. Mathematically demonstrate why this addition logic breaks down completely for complementary events.

Solution Space:

Exercise 3.2.9

A discrete variable can take only four integer states: 1, 2, 3, and 4. The probabilities are given as $P(1) = 0.15$, $P(2) = 0.40$, and $P(3) = 0.25$. Find the probability that the variable takes a state greater than or equal to 3.

- A. 0.20
- B. 0.45
- C. 0.55
- D. 0.80

Scratch Space:

Exercise 3.2.10

An tracking loop captures system fault tags across separate sub-routines. The probability of triggering an Error A profile is m^2 , and the probability of triggering an Error B profile is $3m$. These two errors cannot occur together.

- (a) Express the probability that the system records either Error A or Error B as a polynomial formula in terms of m .
- (b) Given that the combined probability of recording either error evaluates to exactly 0.4, solve for the system metric parameter m .

Solution Space:

Exercise 3.2.11

A fair six-sided die is rolled once. A student wants to find the probability of rolling an even number *or* a multiple of 3. They calculate $P(\text{Even}) = \frac{3}{6}$ and $P(\text{Multiple of 3}) = \frac{2}{6}$, then conclude the answer is $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. This calculation is a failure because:

- A. Dice face properties require multiplication rules.
- B. The events are not mutually exclusive; the score 6 belongs to both sets simultaneously, which means it was double-counted in the addition.
- C. The fractions should be subtracted from unity.
- D. Denominators must scale exponentially.

Scratch Space:

Exercise 3.2.12

A research analyst reviews operational risks in an asset network. The system registers a probability of 0.08 that a network route fails due to hardware dropouts, and a probability of 0.14 that it fails due to software configuration bugs. Software and hardware dropouts are independent but can happen at the same time.

- (a) Discuss why a simple addition of 0.08 and 0.14 fails to give the true probability of a network failure from either cause.
- (b) Explain the logical adjustment required when event boundaries overlap, setting up the concept for non-mutually exclusive sets.

Solution Space:

3.3 Complementary Events

“When the world splits into two outcomes—what we want and everything else—calculating what we don’t want can sometimes lead us to the answer much faster. Shifting our focus to the inverse lets us clear away a mountain of work with a single subtraction.”

Definition: Complementary Events

The complement of an event A , denoted by A' or A^c , represents the collection of all outcomes in the universal sample space S that do not belong to event A . Because an event and its complement are mutually exclusive and completely exhaust the sample space ($n(A) + n(A') = n(S)$), their probabilities satisfy:

$$P(A) + P(A') = 1$$

Theorem: The Law of Complementary Probability

The probability that an event A does not occur is given by subtracting the probability of event A from 1:

$$P(A') = 1 - P(A)$$

Intuitive Core: The total strength of any sample space is always 100%, which maps to exactly 1. If you know there is a 30% chance of rain today (0.30), then there is a 70% chance that it will not rain ($1 - 0.30 = 0.70$). This technique becomes extremely powerful when evaluating phrases like “AT LEAST ONE”, where calculating the single opposite case (“none”) is far easier than summing many separate target tracks. **WARNING: ALWAYS DEFINE THE EXACT EXCLUSION REGION OF THE COMPLEMENT CASE ACCURATELY! THE TRUE INVERSE OF “AT LEAST ONE” IS “NONE AT ALL,” NOT “LESS THAN ALL!”**

Example: Using Complementary Paths for “At Least One” Constraints

A fair coin is tossed 3 times in a row. Find the probability of getting at least one Head across these three sequential tosses.

Detailed Solution Steps:

1. Identify the target event:

Event A = Getting at least one Head

2. Define the absolute complement of this event: The opposite of getting at least one Head is getting *zero* Heads, which means landing on Tails every single time.

$$A' = \{\text{Tails, Tails, Tails}\} \implies \{TTT\}$$

3. Compute the total size of the sample space using exponential counters:

$$n(S) = 2^3 = 8$$

4. Calculate the probability of the complement case:

$$P(A') = P(TTT) = \frac{1}{8}$$

5. Apply the Law of Complementary Probability to find $P(A)$:

$$P(A) = 1 - P(A') = 1 - \frac{1}{8} = \frac{7}{8} = 0.875$$

Hint / Concept Review

If you calculate this directly, you have to find and sum the probabilities for getting exactly 1 Head, exactly 2 Heads, and exactly 3 Heads.

By finding the probability of the single unwanted outcome ($\{TTT\}$) and subtracting it from 1, you reach the same final answer with a fraction of the work.

Subsection 3.3 Practice Exercises**Exercise 3.3.1**

The probability that Owen passes a professional logistics assessment is $\frac{11}{12}$. What is the probability that Owen fails to pass this assessment?

- A. $\frac{1}{12}$
- B. $\frac{1}{11}$
- C. $\frac{1}{2}$
- D. $\frac{11}{12}$

Scratch Space:

Exercise 3.3.2

A quality control inspector examines a batch of components. The probability that a randomly chosen component is defective is 0.04.

- (a) Write out the basic algebraic complement equation used to isolate the non-defective probability space.
- (b) Calculate the probability that a selected component is non-defective.

Solution Space:

Exercise 3.3.3

A box contains 5 red marbles, 6 yellow marbles, and 3 white marbles[cite: 4]. If a student draws one marble at random, what is the probability that the chosen marble is *not* yellow?

- A. $\frac{3}{14}$
- B. $\frac{3}{7}$
- C. $\frac{4}{7}$
- D. $\frac{11}{14}$

Scratch Space:

Exercise 3.3.4

Two fair six-sided dice are rolled simultaneously. Let event E be the outcome that the sum of the scores on the two upper faces is greater than or equal to 4.

- (a) Define the exact complement event E' and list all its specific outcome coordinate pairs.
- (b) Use the complement method to calculate the probability $P(E)$.

Solution Space:

Exercise 3.3.5

The probability that an electronic alarm system triggers correctly during a critical incident is $1 - k^2$. Express the probability that the system fails to trigger during an incident in terms of k .

- A. k
- B. k^2
- C. $1 - k$
- D. $2k - 1$

Scratch Space:

Exercise 3.3.6

An online security platform generates randomized multi-digit passcodes. The probability that a specific passcode track contains zero repeating digits is given by the algebraic expression $\frac{n-4}{n+2}$.

- (a) Construct an algebraic expression for the probability that a passcode contains at least one repeating digit.
- (b) Given that the probability of having at least one repeating digit evaluates to exactly $\frac{1}{3}$, solve for the structural size parameter n .

Solution Space:

Exercise 3.3.7

A fair four-sided spinner labeled 1, 2, 3, and 4 is spun twice in a row. Find the probability that the digit 4 is spun at least once across these two consecutive turns.

- A. $\frac{1}{16}$
- B. $\frac{1}{4}$
- C. $\frac{7}{16}$
- D. $\frac{9}{16}$

Scratch Space:

Exercise 3.3.8

A classroom draft pool contains the records of 18 boys and 12 girls[cite: 4]. The teacher uses an automated system to select a sub-committee of 3 students chosen at random at the same time[cite: 4].

- (a) Formulate the combination ratio needed to find the probability that the selected sub-committee contains only boys, meaning zero girls are selected.
- (b) TRAP WARNING: A student claims that the probability of choosing at least one girl on the committee can be calculated by evaluating $1 - P(\text{all girls})$. Explain why this logic is incorrect, and state the true complementary relationship needed to find the target probability.

Solution Space:

Exercise 3.3.9

A target shooter has a constant success probability of 0.80 of hitting the bullseye on any individual shot. If the shooter fires 3 independent shots at the target, what is the probability that they hit the bullseye at least once?

- A. 0.008
- B. 0.512
- C. 0.992
- D. 0.999

Scratch Space:

Exercise 3.3.10

An experimental test loop tracks signal drops. The probability of zero dropped signals occurring during a 10-minute runtime is given by the variable function $m + 0.15$.

- (a) State the simplified expression for the probability that at least one signal drop occurs during the monitoring runtime.
- (b) Given that the probability of having at least one signal dropout scales to exactly 0.55, find the underlying system parameter m .

Solution Space:

Exercise 3.3.11

A fair coin is tossed 4 times in a row. A student wants to find the probability of getting at least one Tail. They identify the complement as “getting all Heads” ($P = \frac{1}{16}$), but they calculate the final answer as $1 + \frac{1}{16} = \frac{17}{16}$. This calculation is a failure because:

- A. Coin toss calculations must use permutation counters.
- B. The student added the complement probability instead of subtracting it from 1, resulting in a probability value greater than 1, which is structurally impossible.
- C. Complementary logic requires using a base-10 scale.
- D. The inverse probability must be squared.

Scratch Space:

Exercise 3.3.12

An operations analyst reviews structural safety margins for a high-security network server bank. The platform runs 3 independent data links. The probability that any single link drops out due to high traffic is 0.05.

- (a) Discuss why using the complement method is significantly faster than calculating the direct paths when modeling the risk that “at least one link drops out.”
- (b) Complete the full complement calculation steps to prove the exact structural risk probability for the network.

Solution Space:

3.4 Multiplication Law of Probability and Independent Events

“When multiple events unfold in sequence, each stage filters down the remaining possibilities. If these events do not influence one another, their combined strength is found by multiplying their separate individual paths.”

Definition: Independent Events

Two events A and B are said to be independent if the occurrence or non-occurrence of event A has absolutely no effect on the probability of event B occurring. Mathematically, the conditional behavior simplifies completely to its baseline value:

$$P(B \mid A) = P(B)$$

Theorem: The Multiplication Law for Independent Events

If two events A and B are independent, then the probability that *both* event A and event B occur together is the product of their individual probabilities:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Intuitive Core: We apply the Multiplication Law when consecutive conditions are joined by the word “AND”. If the processes are independent (like tossing a coin and rolling a die), the separate paths form a compound grid structure. Multiplying their fractions determines the exact joint probability space. **WARNING: NEVER BLINDLY MULTIPLY PROBABILITIES UNLESS YOU ARE COMPLETELY CERTAIN THE EVENTS ARE INDEPENDENT! IF THE FIRST STAGE ALTERS THE POOL FOR THE SECOND STAGE, YOU MUST ADJUST TO CONDITIONAL MULTIPLICATION!**

Example: Chaining Independent Sequential Stages

A fair coin is tossed and a standard fair six-sided die is rolled once. Find the probability of getting a Head on the coin *and* rolling a 6 on the die.

Detailed Solution Steps:

1. Define the individual event pools:

$$\text{Event } A = \text{Getting a Head} \implies P(A) = \frac{1}{2}$$

$$\text{Event } B = \text{Rolling a 6} \implies P(B) = \frac{1}{6}$$

2. Check for independence: Does landing a Head change the geometric properties or outcomes of the die? No. The events are independent.

3. Apply the Multiplication Law directly:

$$P(A \text{ and } B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{6}$$

$$P(A \text{ and } B) = \frac{1}{12} \approx 0.0833$$

Hint / Concept Review

Identify the keyword *and*. This tells you both conditions must be met at the same time.

Because the coin toss and the die roll cannot influence each other, multiplying their separate probabilities gives you the combined likelihood of this exact path.

Subsection 3.4 Practice Exercises**Exercise 3.4.1**

A fair six-sided die is rolled twice in a row. What is the probability that both rolls turn up a score of 5?

- A. $\frac{1}{36}$
- B. $\frac{1}{18}$
- C. $\frac{1}{6}$
- D. $\frac{1}{3}$

Scratch Space:

Exercise 3.4.2

A student takes a multiple-choice math quiz consisting of 2 independent questions. Each question has 4 choices with exactly one correct answer. The student guesses blindly on both questions.

- (a) State the probability of guessing the correct answer for any single question.
- (b) Calculate the probability that the student guesses the correct answer for *both* questions.

Solution Space:

Exercise 3.4.3

The probability that Owen solves a complex data problem is 0.8, and the probability that his colleague solves the same problem independently is 0.7. Find the probability that *both* of them successfully solve the problem.

- A. 0.15
- B. 0.56
- C. 0.94
- D. 1.50

Scratch Space:

Exercise 3.4.4

A factory runs a two-stage assembly line with independent safety switches. The probability that Switch A operates properly during a power spike is 0.95, and the probability that Switch B operates properly is 0.90.

- (a) Determine the probability that both switches operate properly during a power spike.
- (b) Calculate the probability that both switches fail to operate properly during a power spike.

Solution Space:

Exercise 3.4.5

An asset network has 2 independent tracking nodes. The probability that Node 1 stays online is $1 - x$, and the probability that Node 2 stays online is $2x$. Express the probability that *both* nodes remain online at the same time in terms of x .

- A. $3x - 1$
- B. $2x - 2x^2$
- C. $1 - 2x^2$
- D. $\frac{1-x}{2x}$

Scratch Space:

Exercise 3.4.6

A sequence of 2 independent network tests is run by a server script. The probability of clearing Test 1 is given as $k + 0.2$, and the probability of clearing Test 2 is k .

- (a) Construct an algebraic product mapping the probability that the server clears *both* tests.
- (b) Given that the joint probability of clearing both tests evaluates to exactly 0.24, solve for the structural metric parameter k .

Solution Space:

Exercise 3.4.7

A bag contains 4 red pens and 6 blue pens. A student randomly draws 2 pens one by one *with replacement*. What is the probability that both pens drawn are red?

- A. $\frac{4}{25}$
- B. $\frac{2}{15}$
- C. $\frac{4}{15}$
- D. $\frac{16}{45}$

Scratch Space:

Exercise 3.4.8

An examination center administers a two-part diagnostic test consisting of a written component and a practical track[cite: 1]. The tracking charts show that the probability of a student passing the written test is 0.80, while the probability of passing the practical test is 0.60. Assume student performances across these two tracks are completely independent.

- (a) Evaluate the probability that a randomly selected student clears *both* parts of the exam.
- (b) TRAP WARNING: A test supervisor claims that because the parts are separate, the probability of a candidate passing at least one component can be found by adding the individual rates: $0.80 + 0.60 = 1.40$. Point out the two major flaws in this statement and compute the actual probability of passing at least one component.

Solution Space:

Exercise 3.4.9

A fair coin is tossed 3 times in a row. Find the probability that the sequence lands exactly as Head, then Tail, then Head (HTH).

- A. $\frac{1}{2000}$
- B. $\frac{3}{8}$
- C. $\frac{1}{2}$
- D. $\frac{1}{8}$

Scratch Space:

Exercise 3.4.10

A multi-stage engineering check loops through independent validation sweeps. The probability that an automated scan encounters zero system drops during Sweep 1 is m , and the probability it encounters zero drops during Sweep 2 is $m + 0.1$.

- (a) State the algebraic formula representing the probability that *both* sweeps run without encountering any system drops.
- (b) Given that this joint probability scales to exactly 0.42, evaluate the system parameter m .

Solution Space:

Exercise 3.4.11

The probability that a seed germinates successfully is $\frac{2}{3}$. If 2 such seeds are planted independently, a student wants to find the probability that both seeds germinate and writes down $\frac{2}{3} + \frac{2}{3} = \frac{4}{3}$. This calculation is a failure because:

- A. Germination counts require using combination metrics.
- B. The student added probabilities instead of multiplying them for a joint compound task, resulting in a value greater than 1, which violates probability laws.
- C. Denominators cannot scale linearly.
- D. Independent tracking requires using a negative base.

Scratch Space:

Exercise 3.4.12

An electronics analyst reviews circuit failure logs. Two separate safety channels, Channel X and Channel Y , run in parallel. The probability that Channel X drops out under thermal load is 0.05, and the probability that Channel Y drops out is 0.04. The channels operate independently.

- (a) Discuss how the Multiplication Law models the probability of both safety channels dropping out simultaneously.
- (b) Calculate the probability that the circuit stays safe because at least one channel remains fully operational under thermal load.

Solution Space:

3.5 Conditional Probability and Dependent Events

“When the outcome of the first stage changes the configuration of our world, the next step must adapt to a new reality. Dependent paths require us to pause and recalculate our fractions based on what has already left the pool.”

Definition: Conditional Probability and Dependency

Two events A and B are dependent if the occurrence of event A alters the probability of event B occurring. The probability that event B occurs given that event A has already occurred is called the conditional probability of B given A , denoted by $P(B | A)$, and is defined as:

$$P(B | A) = \frac{P(A \text{ and } B)}{P(A)} \quad (\text{where } P(A) > 0)$$

Theorem: The General Multiplication Law

For any two sequential events A and B , the joint probability that *both* event A and event B occur matches the probability of Stage 1 multiplied by the updated probability of Stage 2 given the outcome of Stage 1:

$$P(A \text{ and } B) = P(A) \times P(B | A)$$

Intuitive Core: Dependent probability typically models sampling *without replacement*. When an object is drawn and kept outside the box, the denominator drops by 1 because the total pool has shrunk. If the item matched your target category, the numerator for that category also drops by 1. We map these dynamic tracking dependencies sequentially using the General Multiplication Law. **WARNING: ALWAYS RE-EVALUATE THE NUMERATOR AND DENOMINATOR FRACTIONS FOR THE SECOND STAGE ASSUMING THAT THE FIRST EVENT HAS COMPLETELY COMPLETED!**

Example: Recalculating Pools Without Replacement

A velvet pouch contains 5 red tokens and 4 blue tokens. A student randomly extracts 2 tokens one after another without replacement. Find the probability that both tokens drawn are red.

Detailed Solution Steps:

1. Define Stage 1: Drawing a red token on the first pick (R_1).

$$P(R_1) = \frac{\text{Red tokens}}{\text{Total tokens}} = \frac{5}{9}$$

2. Define Stage 2 assuming R_1 happened ($R_2 \mid R_1$): One red token is gone, leaving 4 red tokens and 8 total tokens in the pouch.

$$P(R_2 \mid R_1) = \frac{5-1}{9-1} = \frac{4}{8} = \frac{1}{2}$$

3. Apply the General Multiplication Law:

$$P(R_1 \text{ and } R_2) = \frac{5}{9} \times \frac{4}{8} = \frac{20}{72}$$

4. Simplify the resulting tracking fraction:

$$P(R_1 \text{ and } R_2) = \frac{5}{18} \approx 0.278$$

Hint / Concept Review

Notice how the fraction drops from $\frac{5}{9}$ down to $\frac{4}{8}$ for the second draw.

Because the first token was not replaced, it alters the baseline setup for the second stage. Chaining these changing fractions yields the correct joint probability.

Subsection 3.5 Practice Exercises**Exercise 3.5.1**

A box contains 5 red balls, 6 yellow balls, and 3 white balls[cite: 4]. If 2 balls are randomly drawn from the box one after another without replacement, what is the probability that both balls drawn are yellow?

- A. $\frac{15}{18^2}$
- B. $\frac{15}{91}$
- C. $\frac{9}{49}$
- D. $\frac{30}{91}$

Scratch Space:

Exercise 3.5.2

A locker drawer contains 8 white socks, 4 yellow socks, and 2 red socks[cite: 4]. Owen reaches into the drawer and pulls out 2 socks sequentially one by one without replacement[cite: 4].

- (a) Determine the conditional probability that the second sock is white, given that the first sock drawn was red.
- (b) Calculate the joint probability that Owen draws a red sock first followed immediately by a white sock.

Solution Space:

Exercise 3.5.3

A small deck consists of ten cards numbered 2, 3, 5, 8, 11, 11, 12, 15, 19, and 24 respectively[cite: 4]. If 2 cards are drawn sequentially without replacement, find the probability that both cards display odd numbers.

- A. $\frac{1}{33}$
- B. $\frac{2}{9}$
- C. $\frac{5}{18}$
- D. $\frac{25}{100}$

Scratch Space:

Exercise 3.5.4

A class cohort consists of 18 boys and 12 girls[cite: 4]. The teacher selects 2 students randomly one after another without replacement to represent the class at a district seminar.

- (a) Set up a two-stage sequential tree product tracking the outcome path where a boy is chosen first and a girl is chosen second.
- (b) Calculate the probability that the selected representatives consist of exactly 1 boy and 1 girl.

Solution Space:

Exercise 3.5.5

A raffle drum contains n winning tickets and $2n$ blank tickets. If an inspector draws 2 tickets one by one without replacement, express the joint probability that both tickets are winners as an algebraic fraction in terms of n .

- A. $\frac{1}{9}$
- B. $\frac{n-1}{3(3n-1)}$
- C. $\frac{n-1}{9n-3}$
- D. $\frac{n^2}{9n^2-3n}$

Scratch Space:

Exercise 3.5.6

An automated computer network check logs database profiles. A system vault stores $k + 2$ standard asset records and k restricted files. Two records are extracted sequentially without replacement.

- (a) Construct a simplified algebraic expression for the probability that both extracted files are restricted.
- (b) Given that this joint probability tracks out to exactly $\frac{2}{15}$, evaluate the parameter k .

Solution Space:

Exercise 3.5.7

A box contains 4 green pens, 7 blue pens, and 8 black pens[cite: 4]. If 2 pens are randomly selected at the same time, what is the probability that both pens are black?

- A. $\frac{14}{153}$
- B. $\frac{28}{153}$
- C. $\frac{64}{361}$
- D. $\frac{28}{171}$

Scratch Space:

Exercise 3.5.8

An examination board evaluates files for 16 candidate representatives[cite: 4]. The group contains 5 teachers from School A and 11 teachers from School B . The panel extracts a sub-committee of 3 teachers at the same time[cite: 4].

- (a) Explain why selecting objects “at the same time” matches the mathematical properties of sequential random selection without replacement.
- (b) Calculate the probability that the sub-committee contains exactly 2 teachers from School A and 1 teacher from School B .

Solution Space:

Exercise 3.5.9

A coin purse contains 6 silver coins and 4 gold coins. A child shakes out 3 coins one by one without replacement. Find the probability that the child extracts exactly 2 silver coins and 1 gold coin in that exact chronological order (*SSG*).

- A. $\frac{1}{6}$
- B. $\frac{1}{5}$
- C. $\frac{3}{10}$
- D. $\frac{1}{2}$

Scratch Space:

Exercise 3.5.10

A custom network script tracks operational channels. A data block handles m high-priority paths and 6 baseline paths. The monitoring loop samples 2 channels one after another without replacement.

- (a) Formulate the algebraic tracking product representing the probability that the loop samples a baseline channel first followed by a high-priority channel (BH).
- (b) Given that this joint sampling probability evaluates to exactly $\frac{2}{9}$, determine the priority channel parameter m .

Solution Space:

Exercise 3.5.11

A bag contains 6 green marbles and 4 white marbles. A student draws 2 marbles sequentially without replacement. They write down the probability of drawing two green marbles as $\frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = 0.36$. This calculation is a failure because:

- A. Dependent events must be handled using addition.
- B. The student treated the draws as independent, forgetting to decrease both the numerator and the denominator for the second fraction to reflect that a green marble was removed from the pouch. The correct setup is $\frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.
- C. Denominators must scale exponentially.
- D. Sampling marble items requires using permutation bases.

Scratch Space:

Exercise 3.5.12

An operations analyst reviews system dependency logs for a manufacturing plant. A bin stores 12 electronic cards, including 3 cards with firmware errors and 9 functional cards. A testing rig draws a batch of 3 cards sequentially without replacement.

- (a) Discuss how the omission of a replacement step introduces conditional dependency between consecutive trials.
- (b) Compute the explicit probability that the testing rig isolates at least one card containing firmware errors during its 3-card sampling sweep.

Solution Space:

3.6 The General Addition Law

“When two target events overlap, their shared outcomes are counted twice if we simply add their individual paths. To find the true strength of their combination, we must subtract the intersection to balance the ledger.”

Theorem: The General Addition Law

For any two events A and B in a sample space, the probability that event A or event B occurs is given by the sum of their individual probabilities minus the joint probability that both occur simultaneously:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Intuitive Core: We use this law when processing the compound word “OR”, but the underlying categories share an intersection ($P(A \text{ and } B) \neq 0$). For example, if you look for a card that is a Heart or a King, the King of Hearts belongs to both groups. Adding the 13 Hearts and the 4 Kings counts that card twice. Subtracting the 1 shared card fixes the double-counting. **WARNING: NEVER FORGET TO SUBTRACT THE INTERSECTION IF EVENTS ARE NOT MUTUALLY EXCLUSIVE! RE-READ THE RULES CAREFULLY TO CHECK FOR OVERLAPPING OUTCOMES!**

Example: Filtering Out Shared Intersections

A card is drawn at random from a standard deck of 52 playing cards. Find the probability that the drawn card is either a Spade or an Ace.

Detailed Solution Steps:

1. Define the individual event probabilities:

$$\text{Event } A = \text{Drawing a Spade} \implies P(A) = \frac{13}{52}$$

$$\text{Event } B = \text{Drawing an Ace} \implies P(B) = \frac{4}{52}$$

2. Identify the overlapping intersection event (A and B): The card drawn is the Ace of Spades.

$$P(A \text{ and } B) = \frac{1}{52}$$

3. Apply the General Addition Law:

$$P(A \text{ or } B) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52}$$

$$P(A \text{ or } B) = \frac{16}{52} = \frac{4}{13} \approx 0.308$$

Hint / Concept Review

The keyword *or* signals an addition structure.

Because one of the Aces is also a Spade, these two categories are not mutually exclusive. Subtracting $\frac{1}{52}$ removes the double-counted element completely.

Subsection 3.6 Practice Exercises**Exercise 3.6.1**

An integer from 1 to 20 inclusive is selected at random. What is the probability that the chosen number is either an even number *or* a multiple of 5?

- A. $\frac{1}{2}$
- B. $\frac{3}{5}$
- C. $\frac{7}{10}$
- D. $\frac{12}{20}$

Scratch Space:

Exercise 3.6.2

A school database logs the activities of 100 students. The records show that 60 students are members of the Mathematics Club, 45 students are members of the Science Team, and 20 students are registered in both groups. One student is selected at random.

- (a) State why the events “choosing a Mathematics Club member” and “choosing a Science Team member” are not mutually exclusive.
- (b) Calculate the probability that the selected student is a member of either the Mathematics Club *or* the Science Team.

Solution Space:

Exercise 3.6.3

The probability that Owen updates his store's billing app on Monday is 0.40, and the probability that he updates his inventory tracking app is 0.55. If the probability that he completes both updates on Monday is 0.15, find the probability that Owen updates at least one of these apps.

- A. 0.65
- B. 0.80
- C. 0.95
- D. 1.10

Scratch Space:

Exercise 3.6.4

A fair six-sided die is rolled once. Let event A be rolling a number greater than 2, and event B be rolling an even number.

- (a) Write out the individual outcomes for events A , B , and their joint intersection ($A \cap B$).
- (b) Use the General Addition Law formula to calculate $P(A \cup B)$.

Solution Space:

Exercise 3.6.5

Two events X and Y share an overlap. The probability of event X occurring is $3a$, and the probability of event Y is $4a$. If the probability of either event X *or* event Y occurring is $5a$, express the probability of their joint intersection in terms of a .

- A. a
- B. $2a$
- C. $7a$
- D. $12a^2$

Scratch Space:

Exercise 3.6.6

An analyst evaluates transaction paths. The probability that an account experiences a verification delay is k . The probability that it triggers a secure location flag is $k + 0.1$. The probability that a transaction triggers both issues simultaneously is 0.05.

- (a) Construct an algebraic expression for the probability that a transaction encounters either a delay *or* a flag.
- (b) Given that the probability of encountering at least one of these two tracking events is exactly 0.65, solve for the parameter k .

Solution Space:

Exercise 3.6.7

A computer program randomly prints a single letter from the word PROBABILITY. Find the probability that the printed letter is either a vowel *or* one of the letters in the word BOTANY.

- A. $\frac{6}{11}$
- B. $\frac{7}{11}$
- C. $\frac{8}{11}$
- D. $\frac{9}{11}$

Scratch Space:

Exercise 3.6.8

An insurance actuary tracks data for a cohort of 300 drivers. The logs indicate that 180 drivers own a white sedan, 120 drivers own a vehicle equipped with GPS tracking, and 60 drivers own a white sedan that contains GPS tracking. One driver's file is drawn at random.

- (a) Evaluate the probability that the selected file belongs to a driver owning a white sedan *or* a vehicle with GPS tracking.
- (b) TRAP WARNING: A trainee claims that because $180 + 120 = 300$, every single driver in the cohort must own either a white sedan or a GPS-equipped vehicle. Show mathematically why this claim is wrong by finding the exact number of drivers who own neither.

Solution Space:

Exercise 3.6.9

An electronic circuit has two points of vulnerability, A and B . The probability of failure at point A is 0.12, and the probability of failure at point B is 0.08. If the two points fail independently of one another, what is the probability that the circuit experiences a failure at either point A *or* point B ?

- A. 0.1904
- B. 0.2000
- C. 0.2096
- D. 0.9904

Scratch Space:

Exercise 3.6.10

A tracking matrix samples network packages. The probability of encountering a protocol mismatch is m . The probability of encountering a checksum error is $2m$. The two failure conditions occur independently.

- (a) State the algebraic equation for the probability of encountering either a mismatch *or* a checksum error using the independence rule to define the intersection.
- (b) Given that the joint probability of encountering at least one of these errors is exactly 0.44, solve for the system parameter m .

Solution Space:

Exercise 3.6.11

A consumer survey reveals that 70% of households own a laptop, 50% own a tablet, and 30% own both devices. A student wants to find the probability that a household owns either a laptop or a tablet and adds $0.70 + 0.50 = 1.20$. This calculation is a failure because:

- A. Probabilities cannot be converted to percentages.
- B. A probability can never exceed 1; the student failed to subtract the 30% overlap, which double-counted households that own both devices. The correct answer is $0.70 + 0.50 - 0.30 = 0.90$.
- C. Device counts must use combination formulas.
- D. Overlapping systems require exponential multipliers.

Scratch Space:

Exercise 3.6.12

An analyst looks at a dataset where the probability of a target server losing connection due to an external attack is 0.25, and the probability of losing connection due to an internal power drop is 0.10. These two incidents can happen at the same time.

- (a) Explain why the expression $P(A \cup B) = P(A) + P(B)$ overestimates the threat if there is an intersection.
- (b) If the intersection probability is 0.04, calculate the exact probability that the server remains fully connected.

Solution Space:

3.7 Probability with Counting Techniques

“When an event involves selecting groups or arranging lists, writing out paths becomes impossible. We must use our counting keys—permutations and combinations—to build the fractions that govern complex sets.”

Definition: Probability using Combinations and Permutations

When computing the probability of complex configurations, we use P_r^n or C_r^n to establish our structural counters. If the order of selection carries no informational weight, the probability of an event A is given by the ratio of unordered combinations:

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Favorable ways to choose the group using } C_r^n}{\text{Total ways to choose the group from the full pool using } C_r^n}$$

Intuitive Core: This method merges Section 1 counting techniques with Section 3 probability laws. Your strategy must be split into two steps. First, ignore the specific constraint to find the denominator—calculate how many ways the entire group can be chosen or arranged. Second, look at the restriction to find the numerator. **WARNING: MUST MAINTAIN CONSISTENCY ACROSS THE FRACTION! IF YOU USE COMBINATIONS (C_r^n) FOR THE NUMERATOR, YOU MUST USE COMBINATIONS FOR THE DENOMINATOR!**

Example: Using Combinations to Build Probability Ratios

A committee consists of 5 teachers from School A and 4 teachers from School B. If a sub-committee of 3 teachers is randomly selected from the entire committee at the same time, find the probability that the sub-committee contains exactly 2 teachers from School A and 1 teacher from School B.

Detailed Solution Steps:

1. Calculate the denominator $n(S)$: Total ways to choose 3 teachers out of $5 + 4 = 9$ available candidates.

$$n(S) = C_3^9 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$$

2. Calculate the numerator $n(A)$: Choose 2 teachers from School A AND 1 teacher from School B.

$$n(A) = C_2^5 \times C_1^4 = 10 \times 4 = 40$$

3. Combine into the classical ratio:

$$P(A) = \frac{n(A)}{n(S)} = \frac{40}{84} = \frac{10}{21} \approx 0.476$$

Hint / Concept Review

Because the sub-committee is chosen all at once, order does not matter, which means we use C_r^n .

The denominator is the total choice field (C_3^9), and the numerator is the product of our two separate target adjustments ($C_2^5 \times C_1^4$).

Subsection 3.7 Practice Exercises**Exercise 3.7.1**

A box contains 7 red balls and 5 blue balls. If 3 balls are randomly drawn from the box at the same time, what is the probability that all 3 balls drawn are red?

- A. $\frac{7}{44}$
- B. $\frac{21}{220}$
- C. $\frac{35}{220}$
- D. $\frac{7}{12}$

Scratch Space:

Exercise 3.7.2

A tournament director organizes a team pool from a class cohort containing 18 boys and 12 girls. The server randomly picks a temporary group of 4 students from the class at the same time.

- (a) Formulate the combination term needed to determine the total number of ways to pick any 4 students.
- (b) Calculate the probability that the selected group consists of exactly 2 boys and 2 girls.

Solution Space:

Exercise 3.7.3

A rack contains space for 5 keys. Owen wants to hang 3 silver keys and 2 brass keys in a row. If the keys are arranged completely at random, what is the probability that the 2 brass keys sit exactly next to each other?

- A. $\frac{1}{20}$
- B. $\frac{1}{10}$
- C. $\frac{1}{5}$
- D. $\frac{1}{2}$

Scratch Space:

Exercise 3.7.4

A math pool puzzle box contains 6 unique cards labeled A, B, C, D, E , and F . A student arranges all 6 cards in a single straight line.

- (a) Determine the total permutation size of the sample space.
- (b) Use the slot or bundling technique to find the probability that cards A and B are placed next to each other in the line.

Solution Space:

Exercise 3.7.5

A collection contains x functional items and 3 defective items. If an analyst randomly draws 2 items at the same time, express the probability that both chosen items are functional as a fraction in terms of x .

- A. $\frac{x(x-1)}{(x+3)(x+2)}$
- B. $\frac{C_2^x}{C_2^{x+3}}$
- C. $\frac{x^2}{(x+3)^2}$
- D. $\frac{P_2^x}{P_2^{x+3}}$

Scratch Space:

Exercise 3.7.6

An online survey randomly generates system tokens by pulling from a registry containing n available accounts. The software selects a validation group of 2 profiles at the same time.

- (a) Construct an algebraic equation for the probability that a selected pair contains the primary administrator account (assuming there is only 1 primary administrator in the pool of n).
- (b) Given that the probability of including the primary administrator is exactly $\frac{1}{6}$, find the total registry size n .

Solution Space:

Exercise 3.7.7

From a group of 6 boys and 4 girls, a teacher wants to select a project team of 3 students. If the selection is made at random, find the probability that the team contains at least one girl.

- A. $\frac{1}{6}$
- B. $\frac{5}{6}$
- C. $\frac{1}{5}$
- D. $\frac{4}{5}$

Scratch Space:

Exercise 3.7.8

An examination hall coordinator arranges a seating map for 5 candidates taking a Compulsory Math paper and 3 candidates taking an Elective Math module. The 8 students are seated randomly in a single straight row.

- (a) Calculate the total number of unique linear queue permutations for these 8 students.
- (b) TRAP WARNING: A supervisor assumes that because there are more compulsory candidates, the probability that all 3 elective candidates are completely separated can be found using simple division by 3. Use the insertion (gap) method to calculate the true probability of complete separation.

Solution Space:

Exercise 3.7.9

A bag contains 6 white socks, 4 yellow socks, and 2 red socks. If Owen picks 3 socks out of the bag at random without replacement, what is the probability that he draws exactly 1 sock of each color?

- A. $\frac{1}{55}$
- B. $\frac{4}{55}$
- C. $\frac{12}{55}$
- D. $\frac{24}{55}$

Scratch Space:

Exercise 3.7.10

An algorithmic security tool builds configuration codes. The script picks a subset of 3 distinct keys from a resource array containing k alpha characters and 4 numeric tags. The total size of the resource array is $k + 4$.

- (a) State the algebraic formula representing the probability that the tool selects a subset containing only alpha characters.
- (b) Given that the probability of selecting an all-alpha subset evaluates to exactly $\frac{1}{6}$ when the total number of characters is 10, verify the value of k .

Solution Space:

Exercise 3.7.11

A club has 5 male members and 5 female members. A committee of 2 is chosen at random. A student attempts to find the probability that the committee contains 2 women and writes down $\frac{C_2^5}{10} = \frac{10}{10} = 1.0$. This calculation is a failure because:

- A. Combination probabilities cannot evaluate to integers.
- B. The student divided the combination numerator by the raw pool size (10) instead of the total number of ways to choose a committee of 2 ($C_2^{10} = 45$). The correct probability is $\frac{10}{45} = \frac{2}{9}$.
- C. Single-stage picks require exponential counters.
- D. Women cannot be grouped using combination metrics.

Scratch Space:

Exercise 3.7.12

An auditor evaluates data logging errors in an automated inventory register. A box holds 12 security keys. A technician selects a batch of 4 keys at random from the box.

- (a) Discuss how changing the criteria from an ordered sequence to an unordered group transforms your probability tracking from permutations (P_r^n) to combinations (C_r^n).
- (b) If the box contains 8 standard keys and 4 high-priority keys, perform the step-by-step combination ratio math to calculate the probability that the batch contains exactly 2 keys of each type.

Solution Space:

3.8 Probability with Geometric Series

“When a game continues until someone wins, the sequence of turns can theoretically last forever. To sum this infinite chain of diminishing chances, we must link our probability steps to the algebra of an infinite geometric series.”

Definition: Infinite Game Probabilities

Let a sequential game repeat indefinitely until a specific winning condition occurs. If the game consists of a series of independent rounds where the winning probability p remains constant, the probabilities of winning on Turn 1, Turn 2, Turn 3, ... form a descending geometric progression.

Theorem: Sum of an Infinite Geometric Series

The total probability of winning an infinite multi-stage game is found by summing the infinite progression of individual winning paths. If the first term is a and the common ratio is r (where $|r| < 1$), the sum S_∞ evaluates to:

$$S_\infty = \sum_{n=1}^{\infty} a \cdot r^{n-1} = \frac{a}{1-r}$$

Intuitive Core: This technique tracks games that **KEEP GOING**. If Alice and Bob take turns rolling a die, and the first to roll a 6 wins, Alice can win on Turn 1, or on Turn 3 (if Alice fails, Bob fails, then Alice succeeds), or on Turn 5. This chain forms an infinite geometric sequence: $\frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \dots$. **WARNING: INDIVIDUAL TERMS MUST TRACK ALL CUMULATIVE FAILURES THAT CAME BEFORE! EVERY EXTRA TURN REQUIRED TO WIN ADDS FACTORS OF FAILURE MULTIPLIERS TO THE BACKLINE CHIPS!**

Example: Summing Infinite Sequential Turn Spaces

Alice and Bob take turns tossing a fair coin. The first player to toss a Head wins the game. If Alice takes the first turn, find the total probability that Alice wins the game.

Detailed Solution Steps:

1. Map Alice's individual winning turn paths:

$$\text{Win on Turn 1} = P(H) = \frac{1}{2}$$

$$\text{Win on Turn 3} = P(T)P(T)P(H) = \left(\frac{1}{2}\right)^2 \times \frac{1}{2} =$$

$$\text{Win on Turn 5} = P(T)^4 P(H) = \left(\frac{1}{2}\right)^4 \times \frac{1}{2} = \frac{1}{32}$$

2. Identify the geometric progression parameters:

$$\text{First term } a = \frac{1}{2}, \quad \text{Common ratio } r = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

3. Apply the infinite geometric sum formula:

$$P(\text{Alice Wins}) = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{4}} = \frac{\frac{1}{2}}{\frac{3}{4}}$$

$$P(\text{Alice Wins}) = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3} \approx 0.667$$

Hint / Concept Review

Alice only wins on odd turns (1st, 3rd, 5th, ...). For Alice to get a shot on Turn 3, both her 1st shot and Bob's 2nd shot must fail.

This injects a common factor of $P(\text{Fail})^2 = \frac{1}{4}$ into each step. The infinite sum proves Alice holds a 2-to-1 advantage simply by going first.

Subsection 3.8 Practice Exercises**Exercise 3.8.1**

Owen and Tom play a game by taking turns rolling a fair six-sided die. The first player to roll a 6 wins. If Owen rolls first, what is the probability that Owen wins the game on his first or second attempt (Turn 1 or Turn 3)?

- A. $\frac{1}{6}$
- B. $\frac{11}{36}$
- C. $\frac{61}{216}$
- D. $\frac{91}{216}$

Scratch Space:

Exercise 3.8.2

Two players, A and B , take turns rolling a fair four-sided spinner labeled 1 to 4. The first player to land on a 4 wins the game. Player A spins first.

- (a) Write out the first three individual terms of the infinite geometric series tracking Player A 's winning path.
- (b) Calculate the total probability that Player A wins the game.

Solution Space:

Exercise 3.8.3

A computer terminal repeats a high-load database connection test automatically until it clears. The probability of clearing the check on any individual run is 0.40. Find the total probability that the system clears the check on an odd-numbered run (Run 1, 3, 5, ...).

- A. $\frac{5}{8015}$
- B. $\frac{5}{715015}$
- C. $\frac{5}{3151}$
- D. $\frac{25}{29}$

Scratch Space:

Exercise 3.8.4

In an alternating match setup, Owen and his rival fire independent target shots. The probability of hitting the target is 0.30 for Owen and 0.20 for his rival. Owen takes the first shot, and they continue until someone hits the target.

- (a) Deduce the common ratio r that governs the progression of Owen's turn updates by analyzing the joint failures required to return the turn to him.
- (b) Evaluate the total infinite probability that Owen wins the match.

Solution Space:

Exercise 3.8.5

A game has an infinite loop structure. The probability of a player winning on any single turn is p , and the failure rate is $q = 1 - p$. If Player 1 goes first, express the infinite series representing Player 2's total probability of winning the game.

- A. $\frac{q}{1-q^2}$
- B. $\frac{pq}{1-q^2}$
- C. $\frac{p}{1-q^2}$
- D. $\frac{pq}{1-p^2}$

Scratch Space:

Exercise 3.8.6

An automated script tests network routes. The script tests Route A first, then Route B , and loops back and forth until a connection succeeds. The probability of success on any individual attempt is k for Route A and $2k$ for Route B .

- (a) Construct an algebraic expression for the total probability that Route A is the one that establishes the connection using an infinite series sum.
- (b) Given that Route A has an exact total probability of $\frac{5}{11}$ of connecting first, determine the underlying success parameter k .

Solution Space:

Exercise 3.8.7

A fair six-sided die is rolled repeatedly by a single player until a score of either 1 or 2 is obtained. Find the probability that the game ends on a turn that is a multiple of 3 (Turn 3, 6, 9, ...).

- A. $\frac{4}{27}$
- B. $\frac{4}{19}$
- C. $\frac{9}{19}$
- D. $\frac{8}{27}$

Scratch Space:

Exercise 3.8.8

An engineering monitoring system tests circuit arrays sequentially. The system checks automated relays one after another. The probability that any individual relay is faulty is 0.10. The scanning laser evaluates relays continuously down a grid line until it isolates the first faulty unit.

(a) Find the total probability that the first faulty unit is located at an even-numbered position along the grid line (Relay 2, 4, 6, ...).

(b) TRAP WARNING: A technician argues that because the faulty units are distributed randomly, the probability of stopping on an even relay must be exactly equal to stopping on an odd relay ($\frac{1}{2}$). Use geometric series calculus to prove why this claim is wrong.

Solution Space:

Exercise 3.8.9

Three players, A , B , and C , take turns rolling a fair six-sided die in the order $A \rightarrow B \rightarrow C \rightarrow A \dots$. The first player to roll a 6 wins. What is the total probability that Player B wins the game?

- A. $\frac{25}{91}$
- B. $\frac{30}{91}$
- C. $\frac{36}{91}$
- D. $\frac{5}{11}$

Scratch Space:

Exercise 3.8.10

A specialized software script loops through network tokens. The script checks a series of asset channels continuously. The probability of clearing a channel on any single verification sweep is exactly m .

- (a) State the infinite sum equation modeling the total probability that the script finishes its execution exactly on the 4th sweep or later.
- (b) Given that the probability of requiring 4 or more sweeps is exactly 0.512, evaluate the tracking success parameter m .

Solution Space:

Exercise 3.8.11

Two players take turns tossing a coin with a success rate of $\frac{1}{3}$. Player 1 goes first. A student calculates Player 1's total probability of winning by adding the first two turns: $\frac{1}{3} + \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = \frac{13}{27}$. This calculation is a failure because:

- A. Successive turn sets must use permutation bases.
- B. The student only calculated the probability of winning on the 1st or 3rd turn, forgetting that the game can continue to the 5th, 7th, and infinitely many later turns. The full chain must be processed as an infinite geometric series.
- C. Infinite loops can only be evaluated using division factors of ten.
- D. Common ratios must use negative exponents.

Scratch Space:

Exercise 3.8.12

An operations analyst reviews an automated safety script. The script retries an encrypted connection loop indefinitely until a validation key is accepted. The probability of key acceptance on any single retry is 0.15.

- (a) Discuss how the requirement of consecutive failures builds an exponential decay factor into the common ratio of successive turn probabilities.
- (b) Perform the full infinite geometric series calculation to prove that the sum of the probabilities of all possible winning turns across an infinite timeline evaluates to exactly 1, confirming the sample space is closed.

Solution Space:

4 Suggested Answers

Section 1: Counting

Sub 1.1: The Law of Addition

1.1.1 C

1.1.2 (a) The pools must be completely distinct and non-overlapping (vowels and consonants share no common letters).

(b) 26

1.1.3 D

1.1.4 (a) $14 + 9$ (b) 23

1.1.5 C

1.1.6 (a) $4n - 1$ (b) $n = 10$

1.1.7 D

1.1.8 (a) 180

(b) Direct addition overestimates the pool because the 5 overlapping members are counted twice, violating the non-overlapping category rule.

1.1.9 C

1.1.10 (a) $k^2 + k - 2$ (b) $k = 8$

1.1.11 B

1.1.12 (a) It double-counts students who study both subjects.

(b) $\text{Total} = \text{Count}(A) + \text{Count}(B) - \text{Count}(A \text{ and } B)$

Sub 1.2: The Law of Subtraction

1.2.1 B

1.2.2 (a) $\text{Total} = 45 + 30 - 12$ (b) 63

1.2.3 B

1.2.4 (a) 90 (b) 10

1.2.5 A

1.2.6 (a) $n^2 + 6n + 2$ (b) $n = 5$

1.2.7 B

1.2.8 (a) 30 (b) $115 - 30 = 85$ members

1.2.9 B

1.2.10 (a) $m^2 + 6m + 5$ (b) $m = 8$

1.2.11 B

1.2.12 (a) It fails fundamentally if both capture systems record duplicate applications for the same candidate.

(b) Subtract the 350 duplicate forms from the raw sum: $2700 - 350 = 2350$.

Sub 1.3: The Law of Multiplication

1.3.1 C

1.3.2 (a) The structural variations occur sequen-

tially across consecutive independent stages rather than as mutually exclusive choices.

(b) 48

1.3.3 C

1.3.4 (a) 3×12 (b) 36

1.3.5 C

1.3.6 (a) $2n^2 + 5n$ (b) $n = 4$

1.3.7 B[cite: 1]

1.3.8 (a) 80

(b) The number of options in the second stage does not remain constant, rendering direct single-factor independent multiplication invalid.

1.3.9 C

1.3.10 (a) $p^2 + 3p$ (b) $p = 5$

1.3.11 B

1.3.12 (a) It assumes that Stage 2 always provides 7 options regardless of the choice made in Stage 1, introducing a structural error.

(b) $\text{Total} = (1 \times 4) + (4 \times 7) = 32$.

Sub 1.4: The Venn Diagram

1.4.1 B

1.4.2 (a) Left: Robotics exclusive = 40. Center: Intersection = 15. Right: Accounting exclusive = 33.

(b) 12

1.4.3 A

1.4.4 (a) Variable placeholder x in center. (b) 10

1.4.5 A

1.4.6 (a) Map direct expressions to cells. (b) $n = 24$

1.4.7 A

1.4.8 (a) Map metrics to a standard 2-circle layout. (b) 53

1.4.9 B

1.4.10 (a) Assemble standard 3-circle overlapping structure. (b) 25

1.4.11 B

1.4.12 (a) The raw sum exceeds the class population ($25 + 22 + 4 = 51 > 30$), establishing an absolute structural contradiction.

(b) Change parameter: 15 history, 12 geography, 5 neither (overlap = 2).

Sub 1.5: The Tree Diagram**1.5.1** B**1.5.2** (a) Construct standard 2-stage tree. (b) $\{RR, RB, BR, BB\}$ **1.5.3** B**1.5.4** (a) Asymmetrical tree layout cutting off early. (b) 6**1.5.5** B**1.5.6** (a) 2^n (b) $n = 4$ **1.5.7** C**1.5.8** (a) Sketch symmetric 2-stage binary tree. (b) 1 path[cite: 1]**1.5.9** C**1.5.10** (a) Generate 3-stage $2 \times 3 \times 2$ path layout. (b) 12**1.5.11** B**1.5.12** (a) Once the gold coin is removed, zero gold options remain for the next step. (b) Sketch an asymmetrical tree structure.**Sub 1.6: Selection of One Object****1.6.1** C**1.6.2** (a) $\{1, 2, 3, 4, 5, 6\}$ (b) 6**1.6.3** D**1.6.4** (a) 14 (b) 14**1.6.5** C**1.6.6** (a) $n^2 + n - 3$ (b) $n = 4$ **1.6.7** C**1.6.8** (a) 200

(b) Subdivisions do not alter total distinct physical choices available when pulling exactly one worker.

1.6.9 B**1.6.10** (a) 19 (b) $k = 6$ **1.6.11** B**1.6.12** (a) Risk profiles are attributes of a single pool, not independent multi-stage choices. (b) $70 + 30 = 100$.**Sub 1.7: Selection with Replacement****1.7.1** D**1.7.2** (a) Rolling always faces the identical, unshrinking baseline pool size. (b) 216**1.7.3** D**1.7.4** (a) 10^4 (b) 10,000**1.7.5** C**1.7.6** (a) n^2 (b) $n = 12$ **1.7.7** C**1.7.8** (a) 256[cite: 1]

(b) Multi-stage selections form grid configurations rather than mutually exclusive additive paths.

1.7.9 B**1.7.10** (a) p^3 (b) $p = 7$ **1.7.11** B**1.7.12** (a) System A: 216; System B: 81. (b) System A is larger.**Sub 1.8: Selection Without Replacement****1.8.1** B**1.8.2** (a) $12 \times 11 \times 10$ (b) 1320**1.8.3** B**1.8.4** (a) $8 \times 7 \times 6$ (b) 336**1.8.5** C**1.8.6** (a) $n(n-1)$ (b) $n = 11$ **1.8.7** B**1.8.8** (a) 240

(b) One candidate cannot occupy both named leadership roles simultaneously.

1.8.9 C**1.8.10** (a) $(p+1) \times p \times (p-1)$ (b) $p = 5$ **1.8.11** B**1.8.12** (a) Each extra stage multiplies a smaller remaining pool. (b) $6 \times 5 = 30$ vs. $6 \times 5 \times 4 = 120$.**Sub 1.9: Permutation: Factorial Only****1.9.1** D**1.9.2** (a) $4 \times 3 \times 2 \times 1$ (b) 24**1.9.3** B**1.9.4** (a) $7!$ (b) 5040**1.9.5** C**1.9.6** (a) $n!$ (b) $n = 4$ **1.9.7** B**1.9.8** (a) 720

(b) This groups gender clusters abstractly instead of arranging all 6 distinct individuals across the slots.

1.9.9 C**1.9.10** (a) $k!$ (b) $k = 6$ **1.9.11** B**1.9.12** (a) Exponents incorrectly assume duplicate tasks can run concurrently in the timeline. (b) $4! = 24$.**Sub 1.10: Permutation from a Set: nPr** **1.10.1** B**1.10.2** (a) $15P3$ (b) 2730

1.10.3 C**1.10.4** (a) $10 \times 9 \times 8$ (b) 720**1.10.5 C****1.10.6** (a) $n(n-1)(n-2)$ (b) $n = 6$ **1.10.7 B****1.10.8** (a) 11,880

(b) It assumes a flat scalar multiplication, missing the sequential countdown pattern of choice without replacement.

1.10.9 C**1.10.10** (a) $(k+1) \times k$ (b) $k = 7$ **1.10.11 A****1.10.12** (a) Filling 5 slots drives the countdown further down, expanding configuration size. (b) 30 vs. 720.**Sub 1.11: Combination from a Set: C_r^n** **1.11.1 B****1.11.2** (a) C_4^9 (b) 126**1.11.3 B****1.11.4** (a) Combinations; the cups are drawn at the same time and their logging order is entirely irrelevant.

(b) 8008

1.11.5 B**1.11.6** (a) $\frac{n(n-1)}{2}$ (b) $n = 10$ **1.11.7 A****1.11.8** (a) 3060 (b) $C_2^{18} \times C_2^{12} = 10,098$ **1.11.9 B****1.11.10** (a) $C_2^k = \frac{k(k-1)}{2}$ (b) $k = 8$ **1.11.11 B****1.11.12** (a) There is no structural difference; both yield 35 paths.(b) $C_3^7 = C_4^7 = 35$. Selecting 3 elements to include automatically locks 4 elements out.**Sub 1.12: The Law of Division****1.12.1 B****1.12.2** (a) $3! = 6$ (b) 84**1.12.3 B****1.12.4** (a) It divides out the 2 internal sorting variations since structural order carries no label tracking distinction.

(b) 45

1.12.5 C**1.12.6** (a) $\frac{x^2-x}{2}$ (b) $x = 7$ **1.12.7 C****1.12.8** (a) 120

(b) Dividing by 3 cancels a scalar scale, but a 3-

member cluster yields $3! = 6$ layout pathways to filter out.**1.12.9 A****1.12.10** (a) $4! = 24$ (b) 35**1.12.11 A****1.12.12** (a) A scalar divisor fails to eliminate the full cascading network of sorting permutations. (b) 10.**Sub 1.13: The Slot Method****1.13.1 B****1.13.3 C****1.13.5 A****1.13.7 C****1.13.9 B****1.13.11 B****1.13.2** (a) Slot 1 is fixed. Fill remaining 4 empty slots using 4 options. (b) 24**1.13.4** (a) Prioritized stage block ($\square\square$) followed by non-prioritized slots ($\square\square\square$). (b) 48**1.13.6** (a) $(n-1)(n-2)$ (b) $n = 8$ **1.13.8** (a) $4 \times 3 \times 5! = 1440$

(b) Core items cannot be duplicated; after assigning the first slot, the elective/core mix pools must reflect a shrinking base.

1.13.10 (a) $3 \times k \times (k-1) \times (k-2)$ (b) $k = 5$ **1.13.12** (a) Prioritize filling the restricted final slot first to satisfy boundary rules cleanly.(b) Box countdown layout: $4 \times 3 \times 2 \times 1 \times 3 = 72$.**Sub 1.14: The Insertion Method****1.14.1 B****1.14.3 B****1.14.5 B****1.14.7 B****1.14.9 B****1.14.11 B****1.14.2** (a) 7 empty gaps (b) $6! \times P_3^7 = 151,200$ **1.14.4** (a) Consonants: $4!$; Vowels gaps: P_4^5 (b) 2880**1.14.6** (a) $n! \times P_3^{n+1}$ (b) 1440**1.14.8** (a) $5! \times P_3^6 = 14,400$

(b) Subtracting only removes the scenario where all 3 are adjacent, forgetting mixed patterns where only 2 sit together.

1.14.10 (a) $P_2^{k+1} = (k+1) \times k$ (b) $k = 7$ **1.14.12** (a) It maps the spacing parameters directly into localized distinct spaces, avoiding empty adjacent holes. (b) 14,400.

Sub 1.15: Round Table Symmetry

- 1.15.1 B
 1.15.3 A
 1.15.5 A
 1.15.7 B
 1.15.9 B
 1.15.11 B
 1.15.2 (a) Symmetrical rotational shifts do not create distinct new relative configurations.
 (b) 720
 1.15.4 (a) Flipping halves the layout workspace due to mirror reflection equivalence: $\frac{(n-1)!}{2}$. (b) 60
 1.15.6 (a) $(n-1)!$ (b) $n = 7$
 1.15.8 (a) $4! \times 4! = 576$
 (b) Treating the single block yields a circular layout of 5 elements, which has $(5-1)! = 4!$ paths, not $5!$.
 1.15.10 (a) $(m-1)!$ (b) $m = 5$
 1.15.12 (a) Closed loops remove the concept of linear absolute starting nodes. (b) Locking 1 anchor drops options to $4! = 24$.

Section 2: Percentage**Sub 2.1: Basic Changes & Multipliers**

- 2.1.1 C
 2.1.3 C
 2.1.5 C
 2.1.7 B
 2.1.9 B
 2.1.11 B
 2.1.2 (a) 1.40 (b) 450 HKD $\times 1.40 = 630$ HKD
 2.1.4 (a) $\frac{240-320}{320} \times 100\%$ (b) 25% decrease
 2.1.6 (a) $L \times 1.20 = 72$ (b) $L = 60$
 2.1.8 (a) 1020 students
 (b) The baseline changes; a 15% drop on a large base represents a different absolute value than a 15% markup on a small base.
 2.1.10 (a) $\frac{24}{k} \times 100\%$ (b) $k = 200$
 2.1.12 (a) The 10% decrease applies to the newly inflated value, stripping away more absolute magnitude than was added.
 (b) $150 \times 1.10 \times 0.90 = 148.5$ points

Sub 2.2: Profit, Loss, and Discount

- 2.2.1 C
 2.2.3 C
 2.2.5 B
 2.2.7 C
 2.2.9 B
 2.2.11 B
 2.2.2 (a) $S = 250 \times (1 - 0.16)$ (b) 210 HKD
 2.2.4 (a) Marked = 900 HKD; Selling = 720 HKD
 (b) 20% net profit margin
 2.2.6 (a) $M \times (1 - \frac{d}{100})$ (b) $d = 20\%$
 2.2.8 (a) 112% of cost price
 (b) The clearance discount shifts onto a larger marked tag base: $1.60 \times 0.70 = 1.12 \Rightarrow 12\%$ profit.
 2.2.10 (a) $M \times 0.85 = C \times 1.36$ (b) $C = 400$ dollars
 2.2.12 (a) Merely ensure that the initial markup scales significantly higher than the discount adjustment.
 (b) Markup ratio = $\frac{1.05}{0.75} = 1.40 \Rightarrow 40\%$.

Sub 2.3: Successive Changes

- 2.3.1 B
 2.3.3 A
 2.3.5 B
 2.3.7 C

2.3.9 B**2.3.11 B****2.3.2** (a) $50 \times 1.20 \times 0.85$ (b) 51 HKD**2.3.4** (a) 100% of baseline (b) 0% net change**2.3.6** (a) $I \times 1.15 \times (1 + \frac{k}{100}) = 1.38I$ (b) $k = 20$ **2.3.8** (a) 89.6% of Cost Price(b) Discounts apply to consecutively compounding smaller amounts: $1.40 \times 0.80 \times 0.80 = 0.896 \Rightarrow 10.4\%$ net loss.**2.3.10** (a) $W \times 1.08 \times 0.95 = 1.026W$ (b) $W = 100$ **2.3.12** (a) A lower valuation base means a larger percentage rise is required to offset a smaller absolute shock.(b) $0.80 \times 1.25 = 1.00$ (completely unchanged).**Sub 2.4: Compound Interest and Growth****2.4.1 B****2.4.3 B****2.4.4** (a) $15000(1 + r\%)^2 = 17496$ (b) 8%**2.4.5 B****2.4.2** (a) Period rate = 2%; Compounding stages = 4(b) $50000 \times (1.02)^4 - 50000 = 4121.61$ HKD**2.4.6** (a) $(1 + x\%)^3 = \frac{6912}{4000}$ (b) $x = 20\%$ **2.4.7 C****2.4.8** (a) Annual = 112,000 HKD; Monthly = 112,682.50 HKD

(b) Monthly compounding yields an extra 682.50 HKD return because accrued interest is reinvested every month.

2.4.9 C**2.4.10** (a) $P(1 + r\%)^{20} = 2P$ (b) $r = (\sqrt[20]{2} - 1) \times 100$ **2.4.12** (a) No interest has yet accumulated to compound during the initial stage interval.

(b) Simple = 47,200 HKD; Compound = 47,640.64 HKD.

Sub 2.5: Depreciation and Decay**2.5.1 B****2.5.3 B****2.5.5 B****2.5.7 B****2.5.9 B****2.5.11 B****2.5.2** (a) $80000 \times (0.85)^3$ (b) 30,890 HKD loss**2.5.4** (a) $50000(1 - r\%)^2 = 40500$ (b) 10%**2.5.6** (a) $(1 - x\%)^3 = \frac{1024}{2000}$ (b) $x = 20\%$ **2.5.8** (a) $A = 80,000$ HKD; $B = 81,000$ HKD(b) Machine B drops on a shrinking book value base, slowing its absolute loss rate down: Year 3 values are $A = 70,000$ vs. $B = 72,900$.**2.5.10** (a) $P(1 - r\%)^{15} = \frac{1}{2}P$ (b) $r = (1 - \sqrt[15]{0.5}) \times 100$ **2.5.12** (a) Calculating a fixed rate against a smaller book base extracts less magnitude over time.

(b) Year 1: 102,000; Year 2: 86,700; Year 3: 73,695 HKD.

Section 3: Probability**Sub 3.1: Classical Definition****3.1.1 B****3.1.3 B****3.1.5 C****3.1.7 A****3.1.9 B****3.1.11 B****3.1.2** (a) $\{2, 3, 5\}$ (b) $\frac{3}{6} = \frac{1}{2}$ **3.1.4** (a) $\frac{2}{14} = \frac{1}{7}$ (b) $\frac{12}{14} = \frac{6}{7}$ **3.1.6** (a) $\frac{n}{3n-2}$ (b) 13 total ($n = 5$)**3.1.8** (a) $\frac{140}{200} = \frac{7}{10}$

(b) Probability counts relative element density profiles, not the count of abstract label groups.

3.1.10 (a) $\frac{7k}{k^2+7k}$ (b) $k = 4$ **3.1.12** (a) The mechanical system must guarantee equal physical selection opportunities.(b) $\frac{45}{45+5} = \frac{9}{10}$.**Sub 3.2: Addition Law & Mutually Exclusive****3.2.1 C****3.2.3 B****3.2.5 A****3.2.7 C****3.2.9 B****3.2.11 B****3.2.2** (a) A single ball choice can only display exactly one color profile at a time.(b) $\frac{5}{14} + \frac{3}{14} = \frac{4}{7}$ **3.2.4** (a) $0.45 + 0.35 = 0.80$ (b) $1 - 0.80 = 0.20$ **3.2.6** (a) $3k - 0.1$ (b) $k = \frac{7}{30} \approx 0.233$ **3.2.8** (a) $0.05 + 0.11 = 0.16$

(b) Complementary events represent missing space under a single unity ceiling; adding individual inverses overshoots 1.

3.2.10 (a) $m^2 + 3m$ (b) $m = 0.123$

3.2.12 (a) Direct addition double-counts the overlapping scenario where both faults trigger simultaneously.

(b) Subtract the shared intersection: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Sub 3.3: Complementary Events

3.3.1 A

3.3.3 D

3.3.5 B

3.3.7 C

3.3.9 C

3.3.11 B

3.3.2 (a) $P(\text{Safe}) = 1 - P(\text{Defective})$ (b) 0.96

3.3.4 (a) Sum < 4 : $\{(1, 1), (1, 2), (2, 1)\}$ (b) $1 - \frac{3}{36} = \frac{11}{12}$

3.3.6 (a) $\frac{6}{n+2}$ (b) $n = 16$

3.3.8 (a) $\frac{C_3^{18}}{C_3^{30}} = \frac{204}{1015}$

(b) The inverse requires eliminating all girls completely, meaning the target maps to $1 - P(\text{all boys})$.

3.3.10 (a) $0.85 - m$ (b) $m = 0.30$

3.3.12 (a) Evaluating one single complete failure case avoids expanding three separate combination branches.

(b) $1 - (0.05)^3 = 0.999875$.

Sub 3.4: Multiplication Law & Independent

3.4.1 A

3.4.3 B

3.4.5 B

3.4.7 A

3.4.9 A

3.4.11 B

3.4.2 (a) $\frac{1}{4}$ (b) $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

3.4.4 (a) $0.95 \times 0.90 = 0.855$ (b) $0.05 \times 0.10 = 0.005$

3.4.6 (a) $k^2 + 0.2k$ (b) $k = 0.4$

3.4.8 (a) $0.80 \times 0.60 = 0.48$ [cite: 1]

(b) Probability cannot overshoot 1; addition fails for compound independent structures. Correct: $1 - (0.20 \times 0.40) = 0.92$.

3.4.10 (a) $m^2 + 0.1m$ (b) $m = 0.6$

3.4.12 (a) Independent paths multiply risk parameters cleanly: $0.05 \times 0.04 = 0.002$.

(b) $1 - 0.002 = 0.998$.

Sub 3.5: Conditional & Dependent Events

3.5.1 B

3.5.3 B

3.5.5 B

3.5.7 A

3.5.9 A

3.5.11 B

3.5.2 (a) $\frac{8}{13}$ (b) $\frac{2}{14} \times \frac{8}{13} = \frac{8}{91}$

3.5.4 (a) $\frac{18}{30} \times \frac{12}{29} = \frac{36}{145}$ (b) $2 \times \frac{36}{145} = \frac{72}{145}$

3.5.6 (a) $\frac{k(k-1)}{(2k+2)(2k+1)}$ (b) $k = 4$

3.5.8 (a) Extracting all at once prevents drawing identical individual entries across selection slots.

(b) $\frac{C_2^5 \times C_1^{11}}{C_3^{16}} = \frac{110}{560} = \frac{11}{56}$.

3.5.10 (a) $\frac{6m}{m^2+11m+30}$ (b) $m = 4$

3.5.12 (a) Omission of replacement drops the remaining choices pool sequentially.

(b) $1 - P(\text{all clean}) = 1 - \frac{21}{55} = \frac{34}{55}$.

Sub 3.6: The General Addition Law

3.6.1 B

3.6.3 B

3.6.5 B

3.6.7 C

3.6.9 A

3.6.11 B

3.6.2 (a) There are 20 students registered under both groups concurrently.

(b) $\frac{60}{100} + \frac{45}{100} - \frac{20}{100} = \frac{17}{20}$

3.6.4 (a) $A = \{3, 4, 5, 6\}$, $B = \{2, 4, 6\}$, $A \cap B = \{4, 6\}$ (b) $\frac{5}{6}$

3.6.6 (a) $2k + 0.05$ (b) $k = 0.30$

3.6.8 (a) $\frac{180+120-60}{300} = \frac{4}{5}$

(b) The raw count sum overestimates total space due to intersection overlap; $300 - 240 = 60$ own neither.

3.6.10 (a) $3m - 2m^2$ (b) $m \approx 0.163$

3.6.12 (a) Adding base rates double-counts scenarios hitting the intersection.

(b) $1 - (0.25 + 0.10 - 0.04) = 0.69$.

Sub 3.7: Probability with Counting Keys

3.7.1 A

3.7.3 B

3.7.5 B

3.7.7 B

3.7.9 D

3.7.11 B

3.7.2 (a) C_4^{30} (b) $\frac{C_2^{18} \times C_2^{12}}{C_4^{30}} = \frac{136}{369}$

3.7.4 (a) $6! = 720$ (b) $\frac{2! \times 5!}{6!} = \frac{1}{3}$

3.7.6 (a) $\frac{2}{n}$ (b) $n = 12$

3.7.8 (a) $8! = 40,320$

(b) Deploy gap insertion parameters: arrange core (5!), then insert elective into gaps (P_3^6). Prob = $\frac{5}{14}$.

3.7.10 (a) $\frac{C_3^k}{C_3^{k+4}}$ (b) $C_3^k = 20 \implies k = 6$

3.7.12 (a) Stripping away chronological label tracking transforms equations from nPr down to nCr .

(b) $\frac{C_2^8 \times C_2^4}{C_4^{12}} = \frac{168}{495} = \frac{56}{165}$.

Sub 3.8: Probability with Geometric Series

3.8.1 D

3.8.3 B

3.8.5 B

3.8.7 B

3.8.9 B

3.8.11 B

3.8.2 (a) $\frac{1}{4}, \frac{9}{64}, \frac{81}{1024}$ (b) $\frac{\frac{1}{4} - \frac{9}{64}}{1 - \frac{9}{64}} = \frac{4}{7}$

3.8.4 (a) $0.70 \times 0.80 = 0.56$ (b) $\frac{0.30}{1 - 0.56} = \frac{15}{22}$

3.8.6 (a) $\frac{1}{3-2k}$ (b) $\frac{1}{3-2k} = \frac{5}{11} \implies k = 0.4$

3.8.8 (a) $a = 0.09, r = 0.81 \implies \text{Sum} = \frac{9}{19}$

(b) Alternating sequence sweeps introduce systematic head-start advantages, creating an uneven split.

3.8.10 (a) $(1 - m)^3$ (b) $(1 - m)^3 = 0.512 \implies m = 0.2$

3.8.12 (a) Demanding long strings of successive failures scales the common ratio exponentially.

(b) $\frac{p}{1 - (1-p)} = 1.00$ (completely closed universe).